

A circular badge with a dashed border and a solid inner circle, containing the word INDEX in bold capital letters.

INDEX

A rounded rectangular box with a double border, containing the title Digital Logic in a large, bold, sans-serif font.

Digital Logic

1.	Number System and Binary Codes	9.1 – 9.7
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Digital Logic

Syllabus

Boolean algebra. Combinational and sequential circuits. Minimization. Number representations and computer arithmetic (fixed and floating point).

Chapter wise Weightage Analysis

Chapter Paper Year	Ch.1	Ch.2	Ch.3	Ch.4
2008	2	4		
2009	1	1	1	
2010		2	4	2
2011		1	1	5
2012	1	3	1	
2013	1	1	3	
2014 (P1)	1	1	2	
2014 (P2)	3	1		1
2014 (P3)		3	1	2
2015 (P1)		2		3
2015 (P2)		2	2	
2015 (P3)	1	2	2	
2016 (P1)		2	2	3
2016 (P2)		2	1	1
2017 (P1)	1	1	1	2
2017 (P2)	2	4		2
2018	2	3	1	1
2019	2	1	5	
2020	2	2	2	
2021 (P1)	2	2		2
2021 (P2)	4	3		
2022	3		2	
2023	1		3	2

CHAPTER

1

NUMBER SYSTEM AND BINARY CODES

Conversion

1. [NAT] [GATE-2021: 1M]

If x and y are two decimal digits and $(0.1101)_2 = (0.8xy5)_{10}$, the decimal value of $x + y$ is _____.

2. [MCQ] [GATE-2017: 1M]

The representation of the value of a 16-bit unsigned integer X in hexadecimal number system is BCA9. The representation of the value of X in octal number system is

- (a) 136251 (b) 736251
(c) 571247 (d) 571244

3. [NAT] [GATE-2017: 1M]

Consider a quadratic equation $x^2 - 13x + 36 = 0$ with coefficients in a base b . The solutions of this equation in the same base b are $x = 5$ and $x = 6$. Then $b =$ _____.

4. [NAT] [GATE-2015: 2M]

Consider the equation $(43)_x = (y3)_8$ where x and y are unknown. The number of possible solution is _____.

5. [NAT] [GATE-2014: 1M]

The base (or radix) of the number system such that the following equation holds is _____.

$$312/20 = 13.1$$

6. [NAT] [GATE-2014: 1M]

Consider the equation $(123)_5 = (x8)_y$ with x and y as unknown. The number of possible solutions is _____.

7. [MCQ] [GATE-2009: 1M]

$(1217)_8$ is equivalent to

- (a) $(1217)_{16}$ (b) $(028F)_{16}$
(c) $(2297)_{10}$ (d) $(0B17)_{16}$

8. [MCQ] [GATE-2008: 1M]

Let r denote number system radix. The only value(s) of r that satisfy the equation $\sqrt{121_r} = 11_r$ is/are

- (a) decimal 10 (b) decimal 11
(c) decimal 10 and 11 (d) any value > 2

Signed Binary Numbers

9. [MCQ] [GATE-2019: 1M]

In 16-bit 2's complement representation, the decimal number -28 is:

- (a) 1111 1111 0001 1100
(b) 0000 0000 1110 0100
(c) 1111 1111 1110 0100
(d) 1000 0000 1110 0100

10. [NAT] [GATE-2019: 1M]

Two numbers are chosen independently and uniformly at random from the set $\{1, 2, \dots, 13\}$. The probability (rounded off to 3 decimal places) that their 4-bit (unsigned) binary representations have the same most significant bit is _____.

11. [MCQ] [GATE-2019: 1M]

Consider $Z = X - Y$, where X , Y and Z are all in sign-magnitude form. X and Y are each represented in n bits. To avoid overflow, the representation of Z would require a minimum of:

- (a) n bits (b) $n - 1$ bits
(c) $n + 1$ bits (d) $n + 2$ bits

12. [MCQ] [GATE-2018: 2M]

Consider the unsigned 8-bit fixed point binary number representation below, $b_7 b_6 b_5 b_4 b_3 \cdot b_2 b_1 b_0$ where the position of the binary point is between b_3 and b_2 . Assume b_7 is the most significant bit. Some of the decimal numbers listed below cannot be represented exactly in the above representation:

- (i) 31.500 (ii) 0.875 (iii) 12.100 (iv) 3.001

Which one of the following statements is true?

- (a) None of (i), (ii), (iii), (iv) can be exactly represented
(b) Only (ii) cannot be exactly represented
(c) Only (iii) and (iv) cannot be exactly represented
(d) Only (i) and (ii) cannot be exactly represented

13. [MCQ] [GATE-2017:1M]

When two 8-bit numbers $A_7 \dots A_0$ and $B_7 \dots B_0$ in 2's complement representation (with A_0 and B_0 as the least significant bits) are added using a ripple-carry adder, the sum bits obtained are $S_7 \dots S_0$ and the carry bits are $C_7 \dots C_0$.

An overflow is said to have occurred if

- (a) The carry bit C_7 is 1
(b) All the carry bits ($C_7, \dots, C_0$) are 1
(c) $(A_7 \cdot B_7 \cdot \overline{S_7} + \overline{A_7} \cdot \overline{B_7} \cdot S_7)$ is 1.
(d) $(A_0 \cdot B_0 \cdot \overline{S_0} + \overline{A_0} \cdot \overline{B_0} \cdot S_0)$ is 1.

14. [NAT] [GATE-2016:1M]

Consider an eight-bit ripple-carry adder for computing the sum of A and B , where A and B are integers represented in 2's complement form. If the decimal value of A is one, the decimal value of B that leads to the longest latency for the sum to stabilize is _____.

15. [NAT] [GATE-2016:1M]

Let X be the number of distinct 16-bit integers in 2's complement representation. Let Y be the number of distinct 16-bit integers in sign magnitude representation. Then $X - Y$ is _____.

16. [NAT] [GATE-2016: 1M]

The 16-bit 2's complement representation of an integer is 1111 1111 1111 0101; its decimal representation is ____.

17. [MCQ] [GATE-2013: 1M]

The smallest integer that can be represented by an 8-bit number in 2's complement form is

- (a) -256 (b) -128
(c) -127 (d) 0

18. [MCQ] [GATE-2010: 1M]

P is a 16-bit signed integer. The 2's complement representation of P is $(F87B)_{16}$. The 2's complement representation of $8 \times P$ is

- (a) $(C3D8)_{16}$ (b) $(187B)_{16}$
(c) $(F878)_{16}$ (d) $(987B)_{16}$

ANSWER KEY

- | | | | |
|-------------|----------------------|--------------|------------------|
| 1. (3 to 3) | 2. (a) | 3. (8 to 8) | 4. (5 to 5) |
| 5. (5 to 5) | 6. (3 to 3) | 7. (b) | 8. (d) |
| 9. (c) | 10. (0.502 to 0.504) | 11. (c) | 12. (c) |
| 13. (c) | 14. (-1 to -1) | 15. (1 to 1) | 16. (-11 to -11) |
| 17. (b) | 18. (a) | | |

SOLUTIONS

1. (3)

0.5 0.25 0.125 0.0625

$$(0.1101)_2 = ()_{10}$$

$$= 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3} + 1 \times 2^{-4}$$

$$= (0.5 + 0.25 + 0 + 0.625)$$

$$= (0.8125)_{10}$$

$$(0.8125)_{10} = (0.8xy5)_{10}$$

$$x = 1 \quad 4 = 2, \quad x + 4 = 1 + 2 = 3$$



Scan for Video solution



2. (a)

Given X = BCA9 in Hexadecimal format

we know in Binary system B = 1011, C = 1100,

A = 1010, 9 = 1001

$$(BCA9)_{16} \rightarrow (?)_2 \rightarrow (?)_8$$

Therefore, convert hexadecimal number system first in binary then octal system.

$$\begin{array}{ccccccc} 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 3 & 6 & 2 & 5 & 1 & & & & & & & & & & & & \end{array}$$

$$(BCA9)_{16} = (136251)_8$$

Hence correct option is (a).

3. (8)

Given the base of quadratic equation is b.

$$(1)_b x^2 - (13)_b x + (36)_b = (0)_b$$

$$x^2 - (b+3)x + 3b + 6 = 0$$

Let $x = 5$

$$25 - (b+3)5 + 3b + 6 = 0$$

$$25 - 5b - 15 + 3b + 6 = 0$$

$$16 - 2b = 0$$

$$b = 8$$

Let $x = 6$

$$36 - (b+3)6 + 3b + 6 = 0$$

$$36 - 6b - 18 + 3b + 6 = 0$$

$$24 - 3b = 0$$

$$b = 8$$

Hence correct answer is (8).



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4. (5)

$$(43)_x = (y3)_8$$

$$4x + 3 = 8y + 3$$

$$(43)_x = (y3)_8$$

$$x > 4 \quad y < 8$$

$$4x = 8y$$

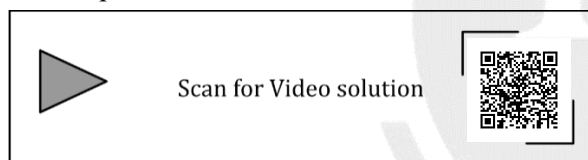
$$x = 2y$$

$$y < 8$$

$$x > 4$$

X	Y	Condition
2	1	(×)
4	2	(✓)
6	3	(✓)
8	4	(✓)
10	5	(✓)
12	6	(✓)
14	7	(✓)

Total 5 possible solution



5. (5)

Let x be the base of the given number

$$\frac{(312)_x}{(20)_x} = (13.1)_x$$

The above number in Decimal will be given as

$$\frac{3 \times x^2 + 1 \times x^1 + 2 \times x^0}{2 \times x^1 + 0 \times x^0} = 1 \times x^1 + 3 \times x^0 + 1 \times x^{-1}$$

$$\frac{3x^2 + x + 2}{2x} = x + 3 + \frac{1}{x}$$

$$\Rightarrow \frac{3x^2 + x + 2}{2x} = \frac{x^2 + 3x + 1}{x}$$

$$\Rightarrow 3x^2 + x + 2 = 2x^2 + 6x + 2$$

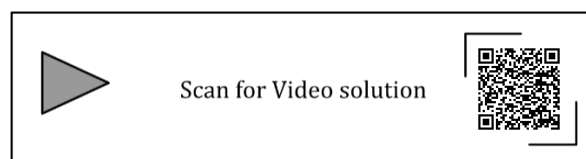
$$\Rightarrow x^2 - 5x = 0$$

$$\Rightarrow (x - 5)x = 0$$

$$x = 0 \quad x = 5$$

Therefore, $x = 5$

Hence, the base of the given number will be 5.



6. (3)

Given equation

$$(123)_5 = (x \ 8)_y$$

By Converting the above equation into decimal form

$$(1 \times 5^2 + 2 \times 5^1 + 3 \times 5^0) = xy + 8$$

$$38 = xy + 8$$

$$xy = 30$$

and $(x < y) (y > 8)$

The possible solutions are

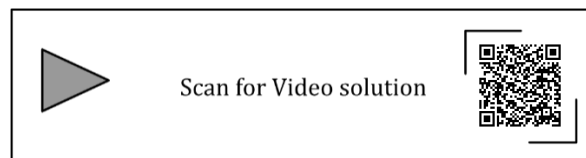
$$x = 1 \quad y = 30 \quad (\checkmark)$$

$$x = 2 \quad y = 15 \quad (\checkmark)$$

$$x = 3 \quad y = 10 \quad (\checkmark)$$

$$x = 5 \quad y = 6 \quad (X)$$

From the above solution only 3 satisfies the given condition.



7. (b)

$$(1217)_8 = ()_{16}$$

$$\swarrow \quad \searrow$$

$$()_2$$

First write the given number in binary representation

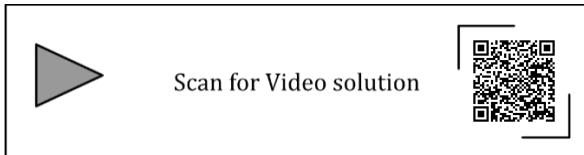
Write each digit 3 bits

$$(1217)_8 = (001010001111)_2$$

Now, to represent the above number in hexadecimal form, make the group of 4 bits.

$$(0010\ 1000\ 1111)_2 = (028F)_{16}$$

Hence, correct answer is option (b).



8. (d)

The given equation is

$$\sqrt{(121)_r} = (11)_r$$

Let r be any base

$$\sqrt{1 \times r^2 + 2 \times r^1 + 1 \times r^0} = 1 \times r^1 + 1 \times r^0$$

$$\sqrt{r^2 + 2r + 1} = r + 1$$

$$\sqrt{(r+1)^2} = r + 1$$

$$\pm(r+1) = r + 1$$

Take negative sign

$$-(r+1) = r + 1$$

$$2r + 2 = 0$$

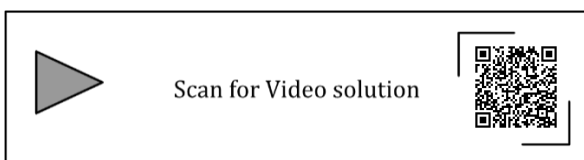
$$r = -1 \text{ (radix cannot be negative)}$$

Take positive sign

$$r + 1 = r + 1$$

for all value which is greater than 2.

Hence, correct answer is option (d).



9. (c)

To find 2's complement of -28 first we convert +28 in binary system and then take 2's complement

$$(28)_{10} = (0000\ 0000\ 0001\ 1100)_2$$

Now taking 2's complement of $(0000\ 0000\ 0001\ 1100)_2 = 1111\ 1111\ 1110\ 0100$

Hence

$$-28 \rightarrow 1111\ 1111\ 1110\ 0100$$

Therefore correct answer is option (C)

10. (0.502)

Decimal	Unsigned representation
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101

From above table 7 number with MSB '0'

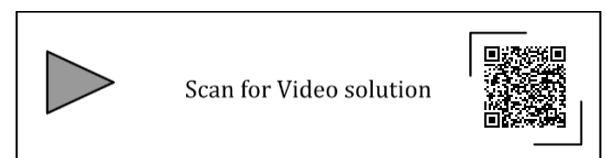
6 number with MSB '1'

Two number are selected independently with same MSB

$$\text{Probability (P)} = P(\text{MSB} = 0) + P(\text{MSB} = 1)$$

$$\text{Probability (P)} = \frac{7}{13} \times \frac{7}{13} + \frac{6}{13} \times \frac{6}{13}$$

$$\text{Probability (P)} = \frac{49}{169} + \frac{36}{169} = 0.5029$$



11. (c)

Overflow occurs when two same sign number are add/sub And result we get in different sign

Example

3 bit sign number

$$(+3) + (1.2) = +5$$

$$\begin{array}{r} 011 \\ + 010 \\ \hline 101 \end{array} \rightarrow \text{in signed it is } -1$$

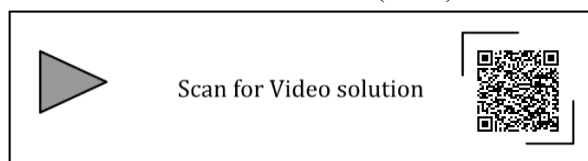
Extension of bit required to avoid overflow

$$Z = X - Y$$

↓ ↓

n-bit n-bit

To avoid overflow z should be (n + 1) bits



12. (c)

Given $b_7 b_6 b_5 b_4 b_3 \cdot b_2 b_1 b_0$

before . 5 bit that means we can represent at max 31_{10}

after . 3 bit that means total 8 combinations are

Binary	Decimal
000	0.000
001	0.125
010	0.250
011	0.375
100	0.500
101	0.625
110	0.750
111	0.875

From above analysis we can't represent $(0.100)_{10}$ and $(0.001)_{10}$

From above analysis only (iii) and (iv) cannot be exactly same represented.

13. (c)

Overflow:

When two same sign number are added and result we will get in different sign.

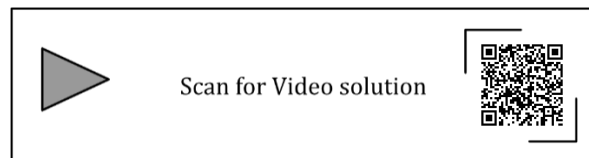
To avoid overflow bit extension required

$$A = A_7 \dots A_0 \quad A_7 = B_7 = 1 \quad S_7 = 0$$

$$B = B_7 \dots B_0 \quad A_7 = B_7 = 0 \quad S_7 = 1$$

$$S = S_7 \dots S_0 \Rightarrow A_7 B_7 \bar{S}_7 + \bar{A}_7 \bar{B}_7 S_7$$

Over flow



14. (-1)

Given:

A = 1 in decimal form

binary form of A = 0000 0001

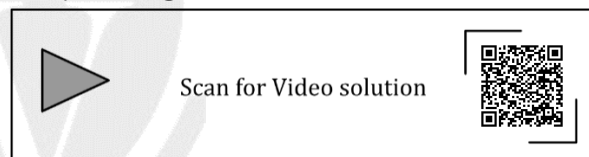
For longest latency, we must have $C_{in} = 1$ at every stage of ripple carry adder.

Maximum delay will be given when we, take B = -1 in decimal and (1111 1111) in binary

$$B = 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \Rightarrow -1$$

$$A = 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \rightarrow +1$$

$$\text{Carry out} = \textcircled{1} \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0$$



15. (1)

If x is in 2's then range of x will be

$$-2^{n-1} \text{ to } 2^{n-1} - 1$$

Here x is 16 bits number, n = 16

$$-2^{16-1} \text{ to } 2^{16-1} - 1$$

$$x = 65536$$

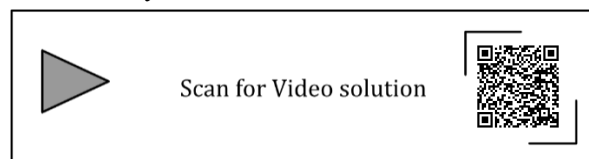
If y is in sign magnitude representation, then range of y will be

$$-(2^{n-1}-1) \text{ to } +(2^{n-1}-1) \quad n = 16$$

$$-(2^{16-1}-1) \text{ to } +(2^{16-1}-1)$$

$$y = 65535$$

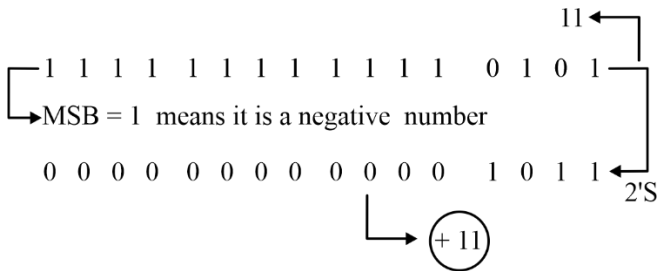
$$\text{Hence, } x - y = 1$$



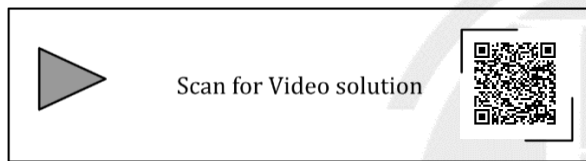
16. (-11)

Given 16-bit number is 1111 1111 1111 0101

Here MSB bit is 1 that means the given number is negative signed number



2's complement of given number is 0000 0000 0000 1011, which is equivalent to + 11 but MSB here is 1 hence number is negative i.e - 11 and correct answer is -11.



17. (b)

For 2's Complement form

The range will be given by

$$-(2^{n-1}) \text{ to } +(2^{n-1} - 1)$$

For $n = 8$ (Given 8 bit number)

The minimum will be

$$\Rightarrow -(2^{8-1})$$

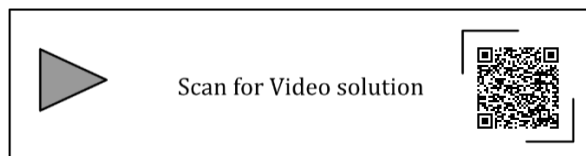
$$\Rightarrow -(2^7)$$

$$\Rightarrow -128$$

The maximum number will be given as

$$(2^{n-1}) = 127$$

Hence, correct answer is option (b).



18. (a)

Method 1:

The given number is

$$P = (F87B)_{16}$$

$$P = 1111100001111011 = (-1925)$$

2's complement of the above number will be given as

$$= 0000011110000101$$

Its decimal representation will be

$$\Rightarrow 1 \times 2^{10} + 1 \times 2^9 + 1 \times 2^8 + 1 \times 2^7 + 4 + 2$$

$$\Rightarrow 1024 + 512 + 256 + 128 + 4 + 1 = 1925$$

Now,

$$8 \times P = 8 \times (-1925) = -15400$$

Its binary representation will be 0011110000101000

Now, 2's complement of the above number will be given as

$$1100001111011000$$

$$8P \rightarrow (C3D8)_{16}$$

Hence, correct answer is option (a)

Method 2:

$$P = (F87B)_{16}$$

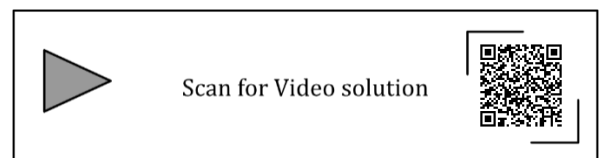
$$= 1111100001111011$$

By multiplying 2^1 , number will be shifted to left by one position, and by multiplying 2^3 , number will be shifted by 3 position to the left.

$$\Rightarrow 11 \ 00 \ 00 \ 1111011000$$

$$8P \Rightarrow (C3D8)_{16}$$

Hence, correct answer is option (a)



BOOLEAN ALGEBRA, LOGIC GATES AND K-MAPS

Boolean Algebra

1. [MCQ] [GATE-2022: 1M]

The dual of a Boolean function $F(x_1, x_2, \dots, x_n, +, \cdot, ')$, written as F^D , is the same expression as that of F with $+$ and $\cdot$ swapped. F is said to be self-dual if $F = F^D$. The number of self-dual functions with n Boolean variables is

- (a) $2n$ (b) 2^{n-1}
(c) 2^{2^n} (d) $2^{2^{n-1}}$

2. [MCQ] [GATE-2017: 1M]

If w, x, y, z are Boolean variables, then which one of the following is INCORRECT?

- (a) $wx + w(x + y) + x(x + y) = x + wy$
(b) $\overline{wx}(y + \overline{z}) + \overline{wx} = \overline{w} + x + \overline{y}z$
(c) $(w\overline{x}(y + x\overline{z}) + \overline{w}\overline{x})y = x\overline{y}$
(d) $(w + y)(wxy + wyz) = wxy + wyz$

3. [MCQ] [GATE-2015: 1M]

The binary operator $\neq$ is defined by the following truth table

P	Q	$p \neq q$
0	0	0
0	1	1
1	0	1
1	1	0

Which one of the following is true about the binary operator $\neq$?

- (a) Both commutative and associative
(b) Commutative but not associative
(c) Not commutative but associative
(d) Neither commutative nor associative

4. [MCQ] [GATE-2015: 1M]

The number of min-terms after minimizing the following Boolean expression is _____.

$$[D' + AB' + A'C + AC'D + A'C'D]'$$

- (a) 1 (b) 2
(c) 3 (d) 4

5. [MCQ] [GATE-2015: 2M]

Given the function $F = P' + QR$, where F is a function in three Boolean variables P, Q and R and $P' = !P$, consider the following statements.

$$(S1) F = \sum(4, 5, 6)$$

$$(S2) F = \sum(0, 1, 2, 3, 7)$$

$$(S3) F = \prod(4, 5, 6)$$

$$(S4) F = \sum(0, 1, 2, 3, 7)$$

Which of the following is true?

- (a) (S1) – False, (S2) – True, (S3) – True, (S4) – False
(b) (S1) – True, (S2) – False, (S3) – False, (S4) – True
(c) (S1) – False, (S2) – False, (S3) – True, (S4) – True
(d) (S1) – True, (S2) – True, (S3) – False, (S4) – False

6. [MCQ] [GATE-2012: 1M]

The truth table

X	Y	$F(X, Y)$
0	0	0
0	1	0
1	0	1
1	1	1

represents the Boolean function

- (a) x (b) $x + y$
(c) $x \oplus y$ (d) y

7. [MCQ] [GATE-2012: 1M]

The amount of ROM needed to implement a 4 bit multiplier is

- (a) 64 bits (b) 128 bits
(c) 1 Kbits (d) 2 Kbits

8. [MCQ] [GATE-2011: 1M]

The simplified SOP (sum of product) form of the boolean expression

$$(P + \bar{Q} + \bar{R}) \cdot (P + \bar{Q} + R) \cdot (P + Q + \bar{R})$$

- (a) $(\bar{P} \cdot Q + \bar{R})$ (b) $(P + Q' \cdot R')$
(c) $(P' \cdot Q + R)$ (d) $(P \cdot Q + R)$

9. [MCQ] [GATE-2011: 1M]

The minterm expansion of $f(P, Q, R) = PQ + Q\bar{R} + P\bar{R}$ is

- (a) $m_2 + m_4 + m_6 + m_7$ (b) $m_0 + m_1 + m_3 + m_5$
(c) $m_0 + m_1 + m_6 + m_7$ (d) $m_2 + m_3 + m_4 + m_5$

10. [MCQ] [GATE-2008: 1M]

If P, Q, R are Boolean variables, then $(P + Q')(P \cdot Q' + P \cdot R)(P' \cdot R' + Q')$ Simplifies to

- (a) $P \cdot \bar{Q}$
(b) $P \cdot \bar{R}$
(c) $P \cdot \bar{Q} + R$
(d) $P \cdot \bar{R} + Q$

Logic Gates

11. [MCQ] [GATE-2022: 1M]

Let, $x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$ where x_1, x_2, x_3, x_4 are Boolean variables, and $\oplus$ is the XOR operator.

Which one of the following must always be TRUE?

- (a) $x_1 x_2 x_3 x_4 = 0$
(b) $x_1 x_3 + x_2 = 0$
(c) $\bar{x}_1 \oplus \bar{x}_3 = \bar{x}_2 \oplus \bar{x}_4$
(d) $x_1 + x_2 + x_3 + x_4 = 0$

12. [MCQ] [GATE-2022: 1M]

Consider the Boolean operator # with the following properties:

$$x \# 0 = x, x \# 1 = \bar{x}, x \# x = 0 \text{ and } x \# \bar{x} = 1$$

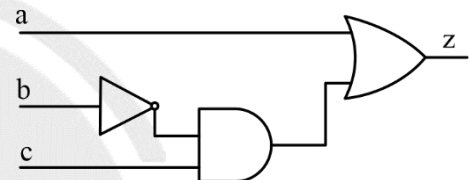
Then $x \# y$ is equivalent to

- (a) $x\bar{y} + \bar{x}y$ (b) $x\bar{y} + \bar{x}\bar{y}$
(c) $\bar{x}y + xy$ (d) $xy + \bar{x}\bar{y}$

13. [MCQ] [GATE-2020: 2M]

Consider the Boolean function $z(a,b,c)$.

Which one of the following minterm lists represents the circuit given above?



- (a) $Z = \Sigma(0, 1, 3, 7)$
(b) $Z = \Sigma(2, 4, 5, 6, 7)$
(c) $Z = \Sigma(1, 4, 5, 6, 7)$
(d) $Z = \Sigma(2, 3, 5)$

14. [MCQ] [GATE-2019: 1M]

Which one of the following is NOT a valid identity?

- (a) $(x + y) \oplus z = x \oplus (y + z)$
(b) $(x \oplus y) \oplus z = x \oplus (y \oplus z)$
(c) $x \oplus y = x + y$, if $xy = 0$
(d) $x \oplus y = (xy + x'y')'$

15. [MCQ] [GATE-2019: 2M]

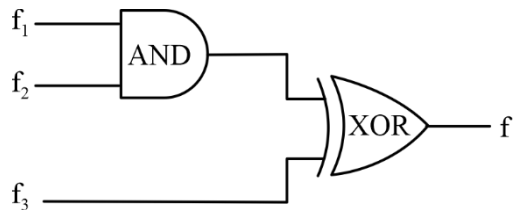
What is the minimum number of 2-input NOR gates required to implement a 4-variable function expressed in sum-of-minterms form as $f = \Sigma(0, 2, 5, 7, 8, 10, 13, 15)$? Assume that all the inputs and their complements are available. _____

16. [MCQ] [GATE-2019: 2M]

Consider three 4-variable functions f_1 , f_2 and f_3 , which are expressed in sum-of-minterms as

$$f_1 = \Sigma(0, 2, 5, 8, 14), \quad f_2 = \Sigma(2, 3, 6, 8, 14, 15), \quad f_3 = \Sigma(2, 7, 11, 14)$$

For the following circuit with one AND gate and one XOR gate, the output function f can be expressed as:



- (a) $\Sigma(2, 14)$
- (b) $\Sigma(7, 8, 11)$
- (c) $\Sigma(2, 7, 8, 11, 14)$
- (d) $\Sigma(0, 2, 3, 5, 6, 7, 8, 11, 14, 15)$

17. [MCQ] [GATE-2018: 1M]

Let $\oplus$ and $\odot$ denote the Exclusive OR and Exclusive NOR operations, respectively. Which one of the following is NOT CORRECT?

- (a) $P \oplus Q = P \odot Q$
- (b) $\bar{P} \oplus Q = P \odot Q$
- (c) $P \oplus \bar{Q} = P \oplus Q$
- (d) $(P \oplus \bar{P}) \oplus Q = (P \odot \bar{P}) \odot \bar{Q}$

18. [MCQ] [GATE-2014: 1M]

Let $\oplus$ denote the Exclusive OR (XOR) operation. Let '1' and '0' denote the binary constants. Consider the following Boolean expression for F over two variables P and Q :

$$F(P, Q) = ((1 \oplus P) \oplus (P \oplus Q)) \oplus ((P \oplus Q) \oplus (Q \oplus 0))$$

The equivalent expression for F is

- (a) $P + Q$
- (b) $(P + Q)'$
- (c) $P \oplus Q$
- (d) $(P \oplus Q)'$

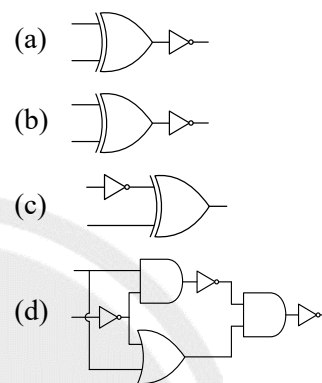
19. [MCQ] [GATE-2013: 1M]

Which one of the following expressions does NOT represent exclusive NOR of x and y ?

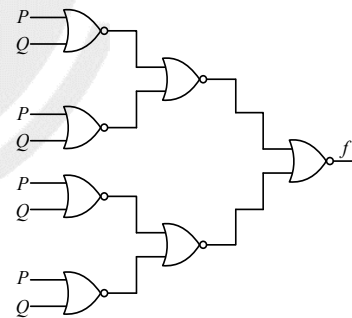
- (a) $xy + x'y'$
- (b) $x \oplus y'$
- (c) $x' \oplus y$
- (d) $x' \oplus y'$

20. [MCQ] [GATE-2011: 1M]

Which one of the following circuits is NOT equivalent to a 2-input X-NOR (exclusive NOR) gate?


21. [MCQ] [GATE-2010: 1M]

What is the Boolean expression for the output f of the combinational logic circuit of NOR gates given below?



- (a) $\overline{Q + R}$
- (b) $\overline{P + Q}$
- (c) $\overline{P + R}$
- (d) $\overline{P + Q + R}$

22. [MCQ] [GATE-2009: 1M]

What is the minimum number of gates required to implement the Boolean function $(AB + C)$ if we have to use only 2-input NOR gates?

- (a) 2
- (b) 3
- (c) 4
- (d) 5

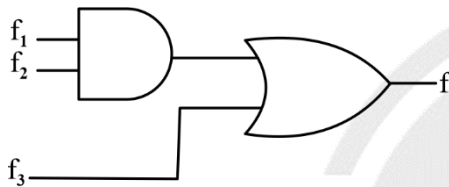
23. [MCQ] [GATE-2008: 1M]

A set of Boolean connectives is functionally complete if all Boolean functions can be synthesized using those. Which of the following sets of connectives is NOT functionally complete?

- (a) EX-NOR (b) implication, negation
(c) OR, negation (d) NAND

24. [MCQ] [GATE-2008: 1M]

Given f_1, f_3 and f in canonical sum of products form (in decimal) for the circuit.



$$F_1 = \sum m(4, 5, 6, 7, 8)$$

$$F_3 = \sum m(1, 6, 15)$$

$$F = \sum m(1, 6, 8, 15)$$

Then f_2 is

- (a) $\sum m(4, 6)$ (b) $\sum m(4, 8)$
(c) $\sum m(6, 8)$ (d) $\sum m(4, 6, 8)$

K-Maps

25. [NAT] [GATE-2022: 1M]

The total number of prime implicants of the function $f(w, x, y, z) = \sum(0, 2, 4, 5, 6, 10)$ is _____.

26. [NAT] [GATE-2018: 2M]

Consider the minterm list form of a Boolean function F given below.

$$F(P, Q, R, S) = \sum m(0, 2, 5, 7, 9, 11) + d(3, 8, 10, 12, 14)$$

Here, m denotes a minterm and d denotes a don't care term. The number of essential prime implicants of the function F is _____.

27. [NAT] [GATE-2017: 1M]

Consider the Karnaugh map given below, where X represents "don't care" and blank represents 0.

ba \ dc	00	01	11	10
00		X	X	
01	1			X
11	1			1
10		X	X	

Assume for all inputs (a, b, c, d) , the respective complements $(\bar{a}, \bar{b}, \bar{c}, \bar{d})$ are also available. The above logic is implemented using 2-input NOR gates only. The minimum number of gates required is _____.

28. [MCQ] [GATE-2017: 1M]

Given $f(w, x, y, z) = \sum m(0, 1, 2, 3, 7, 8, 10) + \sum d(5, 6, 11, 15)$, where d represents the don't-care condition in Karnaugh maps. Which of the following is a minimum product-of-sums (POS) form of $f(w, x, y, z)$?

- (a) $f = (\bar{w} + \bar{z})(\bar{x} + z)$
(b) $f = (\bar{w} + z)(x + z)$
(c) $f = (w + z)(\bar{x} + z)$
(d) $f = (w + \bar{z})(\bar{x} + z)$

29. [MCQ] [GATE-2014: 1M]

Consider the following Boolean expression for F :

$$F(P, Q, R, S) = PQ + P'QR + P'QR'S$$

The minimal sum-of-products form of F is

- (a) $PQ + QR + QSz$
(b) $P + Q + R + S$
(c) $\bar{P} + \bar{Q} + \bar{R} + \bar{S}$
(d) $\bar{P}R + \bar{P}RS + P$

30. [MCQ]

[GATE-2014: 1M]

Consider the following minterm expression for F:

$$F(P, Q, R, S) = \sum 0, 2, 5, 7, 8, 10, 13, 15$$

The minterms 2, 7, 8 and 13 are 'do not care' terms.

The minimal sum-of-products form for F is

- (a) $Q\bar{S} + \bar{Q}S$ (b) $\bar{Q}\bar{S} + QS$
 (c) $\bar{Q}\bar{R}\bar{S} + \bar{Q}R\bar{S} + Q\bar{R}S + QRS$
 (d) $\bar{P}\bar{Q}\bar{S} + \bar{P}QS + PQS + P\bar{Q}\bar{S}$

31. [MCQ]

[GATE-2012: 1M]

What is the minimal form of the karnaugh map shown below? Assume that X denotes a don't care term

ab \ cd	00	01	11	10
00	1	×	×	1
01	×			1
11				
10	1			×

- (a) $\bar{b}\bar{d}$ (b) $\bar{b}\bar{d} + \bar{b}\bar{c}$
 (c) $\bar{b}\bar{d} + \bar{a}\bar{b}\bar{c}\bar{d}$ (d) $\bar{b}\bar{d} + \bar{b}\bar{c} + \bar{c}\bar{d}$

32. [MCQ]

[GATE-2008: 1M]

In the Karnaugh map shown below, X denotes a don't care term. What is the minimal form of the function represented by the Karnaugh map?

ab \ cd	00	01	11	10
00	1	1		1
01	×			
11	×			
10	1	1		×

- (a) $\bar{b}. \bar{d} + \bar{a}. \bar{d}$
 (b) $\bar{a}. \bar{b} + \bar{b}. \bar{d} + \bar{a}. \bar{b}. \bar{d}$
 (c) $\bar{b}. \bar{d} + \bar{a}. \bar{b}. \bar{d}$
 (d) $\bar{a}. \bar{b} + \bar{b}. \bar{d} + \bar{a}. \bar{d}$

33. [MCQ]

[GATE-2008: 1M]

Consider the following Boolean function of four variables $f(A, B, C, D) = \sum(2, 3, 6, 7, 8, 9, 10, 11, 12, 13)$ The function is

- (a) independent of one variable
 (b) independent of two variables
 (c) independent of three variable
 (d) dependent on all the variables

□□□

ANSWER KEY

- | | | | |
|---------|---------|---------|---------|
| 1. (d) | 2. (c) | 3. (b) | 4. (a) |
| 5. (a) | 6. (a) | 7. (d) | 8. (b) |
| 9. (a) | 10. (a) | 11. (c) | 12. (a) |
| 13. (c) | 14. (a) | 15. (3) | 16. (b) |
| 17. (d) | 18. (d) | 19. (d) | 20. (d) |
| 21. (a) | 22. (b) | 23. (a) | 24. (c) |
| 25. (3) | 26. (3) | 27. (1) | 28. (a) |
| 29. (a) | 30. (b) | 31. (b) | 32. (a) |
| 33. (a) | | | |

SOLUTIONS

1. (d)

For $n = 1$, Total number of functions = 4

Total number of self-dual functions = 2

For $n = 2$, Total number of functions = 16

Total number self-dual functions = 4

Now, if we have 'n' variable with us

Then total number of function = 2^{2^n}

Among then number of self-dual functions

$$= 2^{2^{n-1}}$$

Self -Dual:

$$F = F^D$$

$$F = AB$$

$$F^D = A+B$$

$$A \xrightarrow{\text{Dual}} A$$

$$\bar{A} \xrightarrow{\text{Dual}} \bar{A}$$

Ex-

$$F = AB + BC + AC$$

$$F^D = (A + B) \cdot (B + C) (A + C)$$

$$F^D = AB + BC + AC$$

Hence, correct answer is option (d)



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2. (c)

$$\begin{aligned} \text{(a)} \quad wx + w(x+y) + x(x+y) &= x + wy \\ wx + wx + wy + x + xy & \\ x[w + w + 1 + y] + wy & \\ = x + wy & \end{aligned}$$

Therefore, option a is correct

$$\begin{aligned} \text{(b)} \quad \overline{wx}(y + \bar{z}) + \overline{wx} &= \bar{w} + x + \bar{y}z \\ \overline{wx} + \overline{y+z} + \overline{wx} & \\ \bar{w} + x + \bar{y}z + \overline{wx} & \\ \bar{w} + x(1 + \bar{w}) + \bar{y}z &= \bar{w} + x + \bar{y}z \end{aligned}$$

Therefore, option b is correct

$$\begin{aligned} \text{(c)} \quad (\overline{wx}(y + x\bar{z}) + \overline{wx})y &= x\bar{y} \\ \overline{wx}y + w\bar{x}x\bar{z} + \overline{wx}y & \\ [\overline{wx}y + \overline{wx}y] & \end{aligned}$$

$$\bar{x}y(w + \bar{w})$$

$$\bar{x}y$$

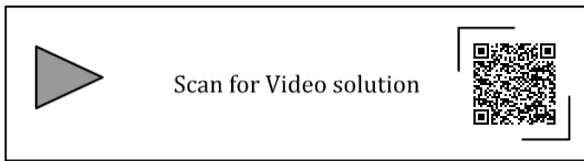
Hence correct answer is option (c)

$$(d) (w + y)(wxy + wyz) = wxy + wyz$$

$$wxy + wyz + wxy + wyz$$

$$wxy + wyz$$

Therefore, option d is correct.



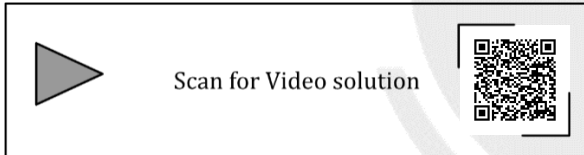
3. (b)

We can see that the given truth table is the output of X - OR gate and we know that X-OR logic follows both commutative and associative operation.

$A \oplus B = B \oplus A \rightarrow$ follow commutative law

$(A \oplus B) \oplus C = A \oplus (B \oplus C)$ Follow associative law.

Hence, correct answer is option (a).



4. (a)

The given Boolean expression is

$$= \overline{\overline{D} + \overline{AB} + \overline{AC} + \overline{ACD} + \overline{ACD}}$$

$$= \overline{\overline{D} + \overline{AB} + \overline{AC} + \overline{CD} (A + \overline{A})}$$

$$= \overline{\overline{D} + \overline{AB} + \overline{AC} + \overline{CD}}$$

$$= \overline{\overline{D} + \overline{CD} + \overline{AB} + \overline{AC}}$$

$$= \overline{(\overline{C} + \overline{D})(\overline{D} + \overline{D}) + \overline{AB} + \overline{AC}}$$

$$\Rightarrow \overline{(\overline{C} + \overline{D}) + (\overline{AB}) + \overline{AC}}$$

$$\Rightarrow \overline{\overline{C} + \overline{D} \cdot \overline{AB} \cdot \overline{AC}}$$

$$\Rightarrow CD[\overline{A} + \overline{B}] \cdot [\overline{A} + \overline{C}]$$

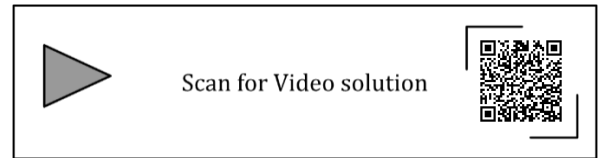
$$\Rightarrow CD[\overline{A} \cdot \overline{A} + \overline{A}\overline{C} + \overline{A}B + \overline{B}\overline{C}]$$

$$\Rightarrow CD[\overline{A}\overline{C} + \overline{A}B + \overline{B}\overline{C}]$$

$$= \overline{A}\overline{C} \cdot CD + ABCD + \overline{B}\overline{C}D$$

$$\Rightarrow ABCD$$

Hence, correct answer is option (a).



5. (a)

Standard minimal form

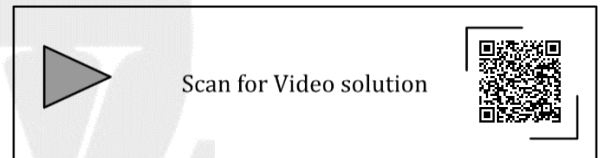
$$F = \overline{P} + QR$$

$$F = \overline{P}(\overline{Q} + Q)(\overline{R} + R) + (\overline{P} + P)QR$$

$$= \overline{P}\overline{Q}\overline{R} + \overline{P}\overline{Q}R + \overline{P}Q\overline{R} + \overline{P}QR + P\overline{Q}\overline{R} + P\overline{Q}R + PQ\overline{R} + PQR$$

$$= \sum m(0, 1, 2, 3, 7)$$

$$= \prod M(4, 5, 6)$$



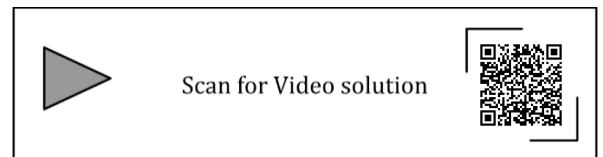
6. (a)

The equation of the given truth table will be given as

$$F(x, y) = x\overline{y} + xy$$

$$= x(\overline{y} + y) = x$$

Hence, correct answer is option (a)



7. (d)

Let us see the 4-bit number multiplication.

$$A = A_3 A_2 A_1 A_0 \longrightarrow 2^4$$

$$B = B_3 B_2 B_1 B_0 \longrightarrow 2^4$$

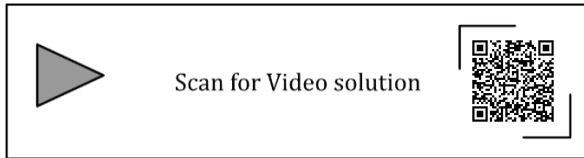
$$A \times B \longrightarrow \overline{\overline{M_7 M_6 M_5 M_4 M_3 M_2 M_1 M_0}}$$

Input = $2^4 \times 2^4$ **output** = 8

The amount of ROM needed to implement a 4 bit multiplier is

$$\begin{aligned}\text{ROM} &= 2^4 \times 2^4 \times 8 \\ &= 2^8 \times 2^3 \\ &= 2^{11} \\ &= 2 \times 2^{10} \\ &= 2 \text{ k bits}\end{aligned}$$

Hence, correct answer is option (d).



8. (b)

The given expression is

$$(P + \bar{Q} + \bar{R}) \cdot (P + \bar{Q} + R) \cdot (P + Q + \bar{R})$$

Let $P + \bar{Q} = x$

$$\Rightarrow (x + \bar{R}) \cdot (x + R) \cdot (P + Q + \bar{R})$$

(By distribution Theorem $x + \bar{R} \cdot R = (x + \bar{R})(x + R)$)

$$\Rightarrow x \cdot (P + Q + \bar{R})$$

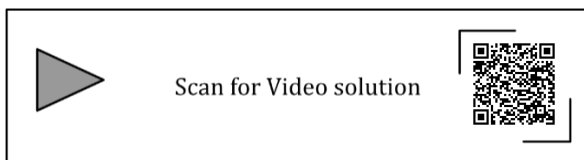
$$\Rightarrow (P + \bar{Q}) \cdot (P + Q + \bar{R})$$

$$\Rightarrow P + PQ + \bar{P}\bar{R} + P\bar{Q} + \bar{Q} \cdot Q + \bar{Q}\bar{R}$$

$$= P[1 + Q + \bar{R} + \bar{Q}] + \bar{Q}\bar{R}$$

$$= P + \bar{Q}\bar{R}$$

Hence, correct answer is option (b).



9. (a)

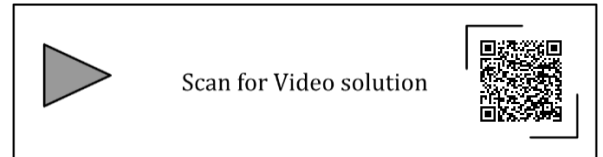
The given expression is

$$F(P, Q, R) = PQ + Q\bar{R} + P\bar{R}$$

The Standard canonical form of the above expression will be given as:

$$\begin{aligned}F(P, Q, R) &= PQ(\bar{R} + R) + (\bar{P} + P)Q\bar{R} + P(\bar{Q} + Q)\bar{R} \\ &= PQ\bar{R} + PQR + \bar{P}Q\bar{R} + PQ\bar{R} + P\bar{Q}\bar{R} + PQR \\ &= m_6 + m_7 + m_2 + m_6 + m_4 + m_6 \\ &= m_2 + m_4 + m_6 + m_7\end{aligned}$$

Hence, correct answer is option (a).



10. (a)

The given expression is

$$(P + Q') (P \cdot Q' + P \cdot R) (P' \cdot R' + Q')$$

$$\Rightarrow (P\bar{Q} + PR + P\bar{Q} + P\bar{Q}R) \cdot (\bar{P}\bar{R} + \bar{Q})$$

$$\Rightarrow (P\bar{Q} + PR) \cdot (\bar{P}\bar{R} + \bar{Q})$$

$$= P(\bar{Q} + R) \cdot (\bar{P}\bar{R} + \bar{Q})$$

$$= (\bar{Q} + R)(P\bar{P}\bar{R} + P\bar{Q})$$

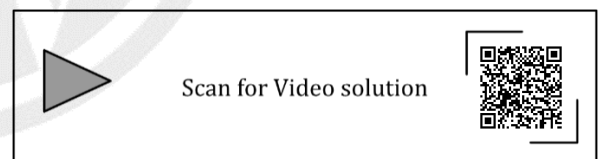
$$= (\bar{Q} + R)(P\bar{Q})$$

$$= P\bar{Q} + P\bar{Q}R$$

$$= P\bar{Q}(1 + R)$$

$$= P\bar{Q}$$

Hence, correct answer is option (a).



11. (c)

We will put $X_1 = 1, X_2 = 1, X_3 = 1, X_4 = 1$

$$1 \oplus 1 \oplus 1 \oplus 1 = 0$$

By taking the above values, we will see all the options one-by-one.

$$\rightarrow X_1 \cdot X_2 \cdot X_3 \cdot X_4 = 1 \cdot 1 \cdot 1 \cdot 1 = 0 \quad (\text{wrong})$$

$$\rightarrow X_1 X_3 + X_2 = 1 \cdot 1 + 1 = 0 \quad (\text{wrong})$$

$$\rightarrow \bar{X}_1 \oplus \bar{X}_3 = \bar{X}_2 \oplus \bar{X}_4 = 0 \oplus 0 = 0 \oplus 0 = 0 \quad (\text{correct})$$

$$\rightarrow X_1 + X_2 + X_3 + X_4 = 0 \quad (\text{wrong})$$

Alternate Method:

$$\text{If } A \oplus B = 0$$

$$\text{Then } A \oplus 0 = B \quad \dots (i)$$

$$\text{Given, } x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$$

$$A = x_1 \oplus x_2, B = x_3 \oplus x_4$$

Substitute values of A, B in equation (i)

$$x_1 \oplus x_2 \oplus 0 = x_3 \oplus x_4$$

$$x_1 \oplus x_2 = x_3 \oplus x_4$$

$$\text{We know that } \overline{A \oplus B} = \overline{A} \oplus \overline{B}$$

$$\overline{x_1 \oplus x_2} = \overline{x_3 \oplus x_4}$$

$$\overline{x_1} \oplus \overline{x_2} = \overline{x_3} \oplus \overline{x_4}$$

Hence, correct answer is option (c)



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**12. (a)**

Given

$$x \# 0 = x, \quad x \# 1 = \bar{x},$$

$$x \# x = 0, \quad x \# \bar{x} = 1$$

We can see from the given condition that the logic operator # resembles the logic of EX-OR gate.

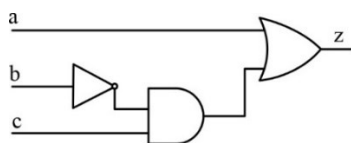
Therefore,

$$x \# y = \bar{x}y + x\bar{y}$$

Hence correct answer is option (a)



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**13. (c)**

$$z = a + \bar{b}c$$

standard canonical form

$$z = a(\bar{b} + b)(\bar{c} + c) + (\bar{a} + a)\bar{b}c$$

$$z = a\bar{b}\bar{c} + a\bar{b}c + ab\bar{c} + abc + \bar{a}\bar{b}c + \bar{a}b\bar{c}$$

$$z = \sum m(1, 4, 5, 6, 7)$$

Hence correct answer is option (c).



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**14. (a)****Option (a)**

$$(x + y) \oplus z = x \oplus (y + z)$$

$$(x + y) \oplus z = (x + y) \bar{z} + (\overline{x + y}) z$$

$$(x + y) \oplus z = x\bar{z} + y\bar{z} + \bar{x}\bar{y}z$$

$$x \oplus (y + z) = x(\overline{y + z}) + \bar{x}(y + z)$$

$$x \oplus (y + z) = x\bar{y}\bar{z} + \bar{x}y + \bar{x}z$$

$$(x + y) \oplus z \neq x \oplus (y + z)$$

Option (b)

$$(x \oplus y) \oplus z = x \oplus (y \oplus z)$$

X-OR gate follows associative law

Option (c)

$$x \oplus y = x + y, \text{ if } x \cdot y = 0$$

$$x \oplus y = \bar{x}y + x\bar{y}$$

$$= (x + y)(\bar{x} + \bar{y})$$

$$= (x + y) = 1 = x + y$$

$$\bar{x}\bar{y} = 1$$

$$\bar{x} + \bar{y} = 1$$

Option (d)

$$x \oplus y = (xy + \bar{x}\bar{y})$$

$$x \oplus y = \bar{x} \odot \bar{y}$$

$$x \oplus y = \overline{xy} + \bar{x}\bar{y}$$

Hence option (a) is correct answer.

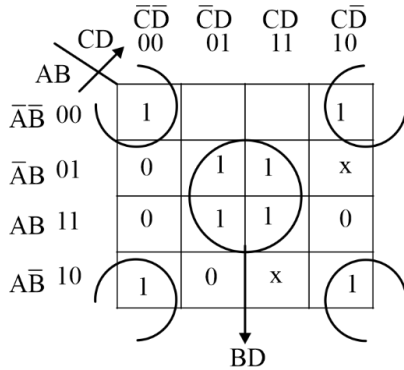


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15. (3)

$$F = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$$



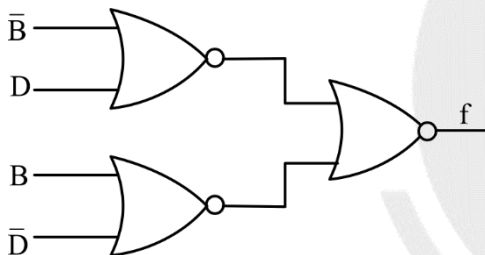
Simplified expression from above K-Map

In SOP

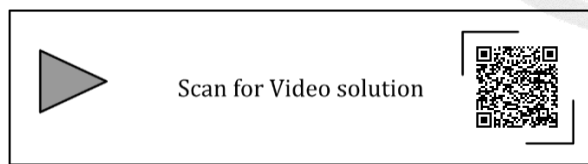
$$F = \bar{B}\bar{D} + BD$$

In POS

$$F = (\bar{B} + D)(B + \bar{D})$$



From above logic circuit minimum number NOR gate required is 3.



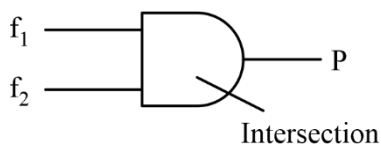
16. (b)

Given

$$f_1 = \sum(0, 2, 5, 8, 14)$$

$$f_2 = \sum(2, 3, 6, 8, 14, 15)$$

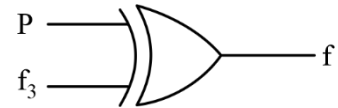
$$f_3 = \sum(2, 7, 11, 14)$$



$$P = f_1 \cap f_2$$

$$P = (0, 2, 5, 8, 14) \cap (2, 3, 6, 8, 14, 15)$$

$$P = (2, 8, 14)$$



$$f = P \oplus f_3$$

$$f = P \bar{f}_3 + \bar{P} f_3$$

$$\bar{P} f_3 = (2, 8, 14) \cdot (0, 1, 3, 5, 6, 8, 9, 10, 12, 13, 15)$$

$$P \bar{f}_3 = (8)$$

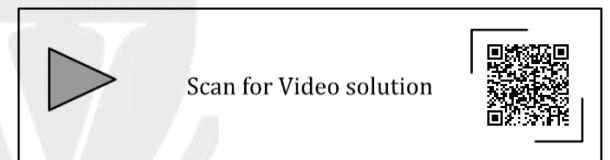
$$\bar{P} f_3 = (0, 1, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 15) \cdot (2, 7, 11, 14)$$

$$\bar{P} f_3 = (7, 11)$$

$$f = (8) \oplus (7, 11)$$

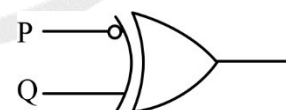
$$f = (7, 8, 11)$$

$$f = \sum(7, 8, 11)$$



17. (d)

Option (b)

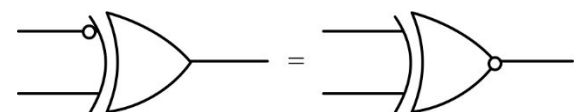


$$f = \bar{P} \oplus Q$$

$$\text{Let } \bar{P} = Z$$

$$f = Z \oplus Q = \bar{Z}Q + Z\bar{Q}$$

$$f = \bar{\bar{P}}Q + \bar{P}\bar{Q} = PQ + \bar{P}\bar{Q} = P \odot Q$$



Hence option (b) is correct

Option (c)

$$\begin{aligned}\bar{P} \oplus \bar{Q} &= \bar{P} \bar{Q} + \bar{P} Q \\ &= P\bar{Q} + \bar{P} Q \\ &= P \oplus Q\end{aligned}$$

Hence option (c) is correct

Option (d)

$$(P \oplus \bar{P}) \oplus Q = (P \odot \bar{P}) \odot \bar{Q}$$

$$1 \oplus Q = \bar{Q} \neq 0 \odot \bar{Q} = Q$$

Hence option (d) is incorrect.



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$$= ((x' + y)(x + y'))$$

$$= x'y' + xy$$

$$\rightarrow xy + x'y' = x \odot y \quad [\text{Exclusive OR}]$$

$$\rightarrow x \oplus y' = x \odot y \quad [\text{Exclusive OR}]$$

$$\rightarrow x' \oplus y = x \odot y \quad [\text{Exclusive OR}]$$

$$\rightarrow x' \oplus y' = x \oplus y \quad [\text{Exclusive NOR}]$$

Hence, correct answer is option (d).



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18. (d)

$$\text{Let } P \oplus Q = Z$$

$$F = (1 \oplus P) \oplus (Z) \oplus (Z) \oplus (Q \oplus 0)$$

$$= [\bar{P} \oplus P \oplus Q] \oplus [P \oplus Q \oplus Q]$$

$$= [1 \oplus Q] \oplus [P \oplus 0]$$

$$= \bar{Q} \oplus P$$

$$= \overline{P \oplus Q}$$

$$\text{Note: } \begin{cases} P \oplus 1 = \bar{P} \\ Q \oplus 0 = Q \\ \bar{P} \oplus Q = P \odot Q = \overline{P \oplus Q} \\ Z \oplus Z = 0 \end{cases}$$

Hence, correct answer is option (d).



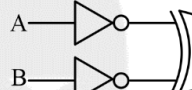
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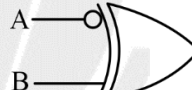


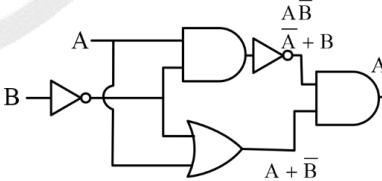
20. (d)

Let us see the output of all the given options we get

(a)  $\overline{A \oplus B} = A \odot B$

(b)  $\overline{A \oplus B} = A \odot B$

(c)  $\overline{A \oplus B} = A \odot B$

(d)  $\overline{A \oplus B} = A \odot B$

Hence, correct answer is option (d).



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19. (d)

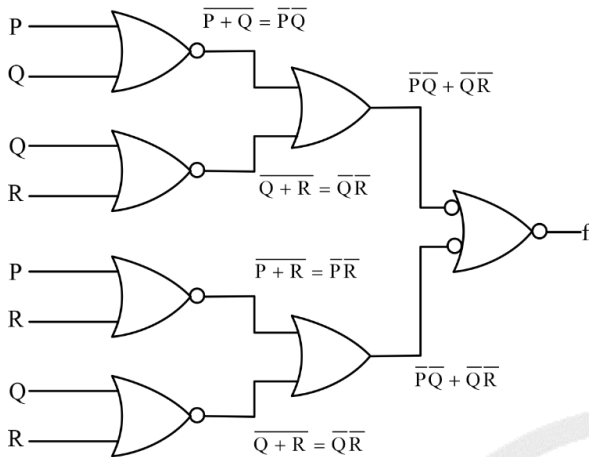
We know that Exclusive NOR of x and y

$$= (\overline{x \oplus y})$$

$$= (\overline{xy' + x'y})'$$

21. (a)

Let us see the output of the given logic circuit at each stage, we get



The output of the above logic circuit will be given by

$$f = (\overline{PQ} + \overline{QR})(\overline{PR} + \overline{QR})$$

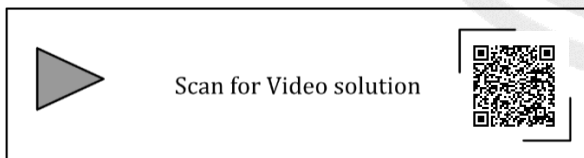
$$f = \overline{PQR} + \overline{PQR} + \overline{PQR} + \overline{QR}$$

$$f = \overline{PQR} + \overline{QR}$$

$$f = \overline{QR} (\overline{P} + 1)$$

$$f = \overline{QR}$$

Hence, correct answer is option (a).



22. (b)

The given expression is

$$f = (AB + C)$$

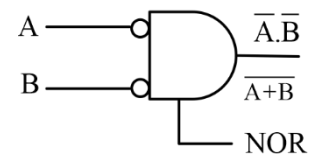
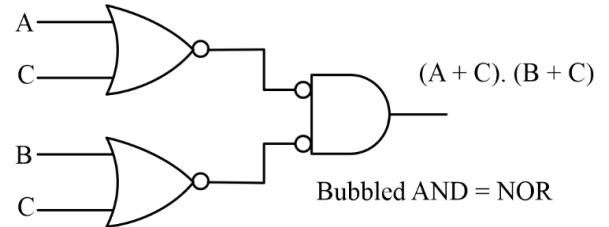
By using distribution theorem, we can write above expression as

$$f = (A + C) \cdot (B + C)$$

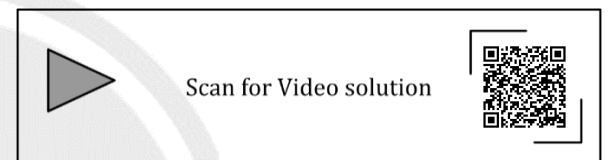
Note: Whenever in the problem, NOR gates are asked, write the function in POS.

Now, the above expression is in POS form.

Hence, three NOR gates will be required to implement f as shown below



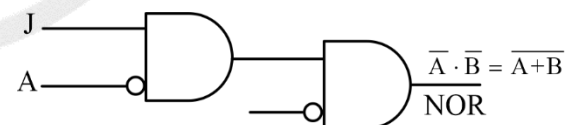
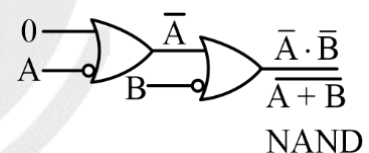
Hence, correct answer is option (b).



23. (a)

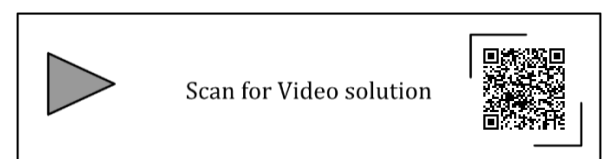
We now, that

NAND/NOR is functionally complete.

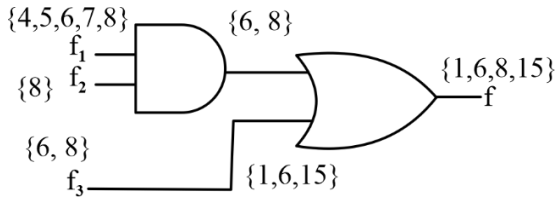


And from the above options only EX-NOR is not functionally complete

Hence, correct answer is option (a).



24. (c)



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25. (3)

The given function is

$$f(w,x,y,z) = \Sigma(0, 2, 4, 5, 6, 10)$$

The k-map of the above function will be given as below

yz \ wx	00	01	11	10
00	1			1
01	1	1		1
11				
10				1

From the above k-map

$$F = \overline{W}\overline{Z} + \overline{W}X\overline{Y} + \overline{X}Y\overline{Z}$$

 $\therefore$ Total prime implicants = 3

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26. (3)

K-Map of given boolean function F

	RS	$\overline{R}\overline{S}$ 00	$\overline{R}S$ 01	RS 11	$R\overline{S}$ 10
PQ					
$\overline{P}\overline{Q}$ 00		1		x	1
$\overline{P}Q$ 01		0	1	1	
PQ 11		x			x
$P\overline{Q}$ 10		x	1	1	x

From above K-Map number of Essential prime implicants is 3.



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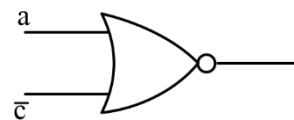
27. (1)

The given k-map is shown below

ba \ dc	00	01	11	10
00		x	x	
01	1			x
11	1			1
10		x	x	

here only one quad group is possible then therefore, representation of given k map is

$$F = \overline{a}c = \overline{a + \overline{c}} = \overline{a}c$$



Hence only one 2-input NOR gate required.



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28. (a)

$$f(w, x, y, z) = \sum m(0, 1, 2, 3, 7, 8, 10) + \sum d(5, 6, 11, 15)$$

yz \ wx					
		00	01	11	10
w+x	00	1	1	1	1
w+x̄	01	0	x	1	x
̄w+x̄	11	0	0	x	0
̄w+x	10	1	0	x	1

$(\bar{x} + z)$ (pointing to row 01)
 $(\bar{w} + \bar{z})$ (pointing to column 11)

from above K-Map 2 quad is possible

$$\text{Therefore } f(w, x, y, z) = (\bar{x} + z)(\bar{w} + \bar{z})$$

Hence option answer is option (a).



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29. (a)

The given function is

$$F(P, Q, R, S) = PQ + \bar{P}Q\bar{R}S$$

Its standard canonical form will be

$$F(P, Q, R, S) = PQ(\bar{R} + R)(\bar{S} + S) + \bar{P}QR(\bar{S} + S) + \bar{P}Q\bar{R}S$$

$$F(P, Q, R, S) = PQ(\bar{R} + R)(\bar{S} + S) + \bar{P}QR(\bar{S} + S) + \bar{P}Q\bar{R}S$$

$$= PQR\bar{S} + PQ\bar{R}S + PQR\bar{S} + PQRS + \bar{P}QR\bar{S} + \bar{P}QRS + \bar{P}QR\bar{S}$$

$$\sum m = (12, 13, 14, 15, 6, 7, 5)$$

Now, the k-map of the above function will be given as below

RS \ PQ		00	01	11	10
00					
01			1	1	1
11		1	1	1	1
10					

$$= PQ + QS + QR$$

Hence, correct answer is option (a).



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30. (b)

The given function is

$$F(P, Q, R, S) = \sum 0, 2, 5, 7, 8, 10, 13, 15$$

The k-map of the above functions is given below

RS \ PQ		00	01	11	10
00		1			x
01			1	x	
11			x	1	
10		x			1

From the above k-map

$$F = \bar{Q}\bar{S} + QS$$

Hence, correct answer is option (b).



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31. (b) The given k-map is as shown below

cd \ ab		00	01	11	10
00		1	X	X	1
01		X			1
11					
10		1			X

From the given k-map, function f will be given as

$$f = \bar{b}\bar{c} + \bar{b}\bar{d}$$

Hence correct answer is option (b).



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32. (a)

Given truth table is

ab \ cd	00	01	11	10
00	1	1		1
01	x			
11	x			
10	1	1		x

From the above k-map the function f will be given as

$$f = \bar{a}\bar{d} + \bar{b}\bar{d}$$

Hence, correct answer is option (a).



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33. (a)

The given expression is

$$f(A, B, C, D) = \sum m(2, 3, 6, 7, 8, 9, 10, 11, 12, 13)$$

k-map of the above expression will be given as

AB \ CD	00	01	11	10
$\bar{A}\bar{B}$ 00			1	1
$\bar{A}B$ 01			1	1
AB 11	1	1		
$A\bar{B}$ 10	1	1	1	1

AB \ CD	00	01	11	10
$\bar{A}\bar{B}$ 00			1	1
$\bar{A}B$ 01			1	1
AB 11	1	1		
$A\bar{B}$ 10	1	1	1	1

$$f = \bar{A}\bar{C} + A\bar{C} + A\bar{B}$$

$$f = \bar{A}\bar{C} + A\bar{C} + \bar{B}C$$

We can see from the above expression that f is independent of D .

Hence, correct answer is option (a).



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□□□

BCD Adder and Subtractor

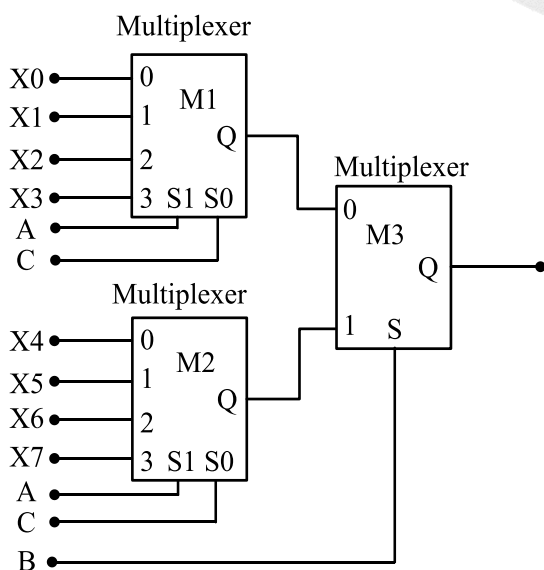
1. [NAT] [GATE-2016: 1M]

Consider an eight-bit ripple-carry adder for computing the sum of A and B where A and B are integers represented in 2's complement form. If the decimal value of A is one, the decimal value of B that leads to the longest latency for the sum to stabilize is_____.

Multiplexer

2. [MCQ] [GATE-2023: 2M]

A Boolean digital circuit is composed using two 4-input multiplexers (M1 and M2) and one 2-input multiplexer (M3) as shown in the figure. X0-X7 are the inputs of the multiplexers M1 and M2 and could be connected to either 0 or 1. The select lines of the multiplexers are connected to Boolean variables A, B and C as shown.



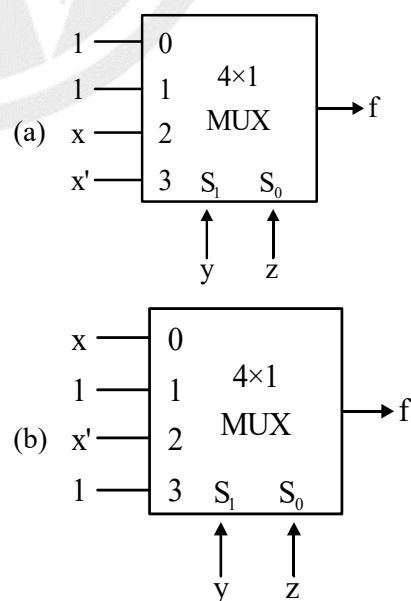
Which one of the following set of values of (X0, X1, X2, X3, X4, X5, X6, X7) will realise the Boolean function $\bar{A} + \bar{A} \cdot \bar{C} + A \cdot \bar{B} \cdot C$?

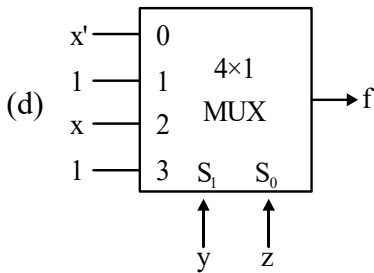
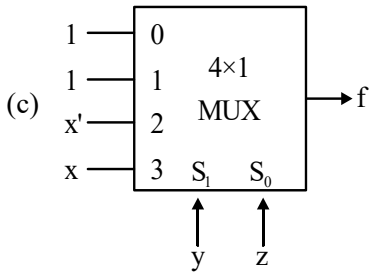
- (a) (1, 1, 0, 0, 1, 1, 1, 0)
- (b) (1, 1, 0, 0, 1, 1, 0, 1)
- (c) (1, 1, 0, 1, 1, 1, 0, 0)
- (d) (0, 0, 1, 1, 0, 1, 1, 1)

3. [MCQ] [GATE-2021: 1M]

Which one of the following circuits implements the Boolean function given below?

$f(x, y, z) = m_0 + m_1 + m_3 + m_4 + m_5 + m_6$ where m_i is the i^{th} minterm.



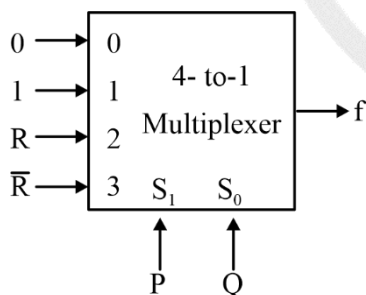


4. [NAT] [GATE-2020: 1M]

A multiplexer is placed between a group of 32 registers and an accumulator to regulate data movement such that at any given point in time the content of only one register will move to the accumulator. The number of select lines needed for the multiplexer is _____.

5. [MCQ] [GATE-2014: 2M]

Consider the 4-to-1 multiplexer with with two select S_1 and S_0 given below.

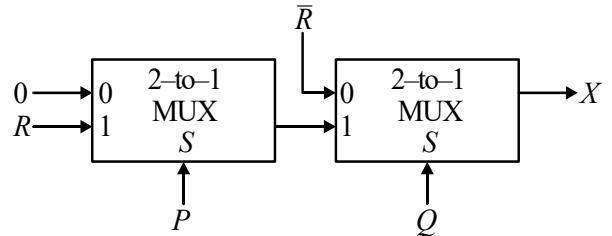


The minimal sum of products from the Boolean expression for the output F of the multiplexer is-

- (a) $\bar{P}Q + Q\bar{R} + P\bar{Q}R$
 (b) $\bar{P}Q + \bar{P}Q\bar{R} + PQR + P\bar{Q}R$
 (c) $\bar{P}QR + \bar{P}Q\bar{R} + Q\bar{R} + P\bar{Q}R$
 (d) $PQ\bar{R}$

6. [MCQ] [GATE-2016: 1M]

Consider the two cascaded 2 to 1 multiplexers as shown in the figure.



The minimal sum of products form of the output X is

- (a) $\bar{P}\bar{Q} + PQR$ (b) $\bar{P}Q + QR$
 (c) $PQ + \bar{P}\bar{Q}R$ (d) $\bar{Q}\bar{R} + PQR$

7. [MCQ] [GATE-2014: 1M]

Consider the following combinational function block involving four Boolean variables x, y, a, b where x, a, b are inputs and y is the output.

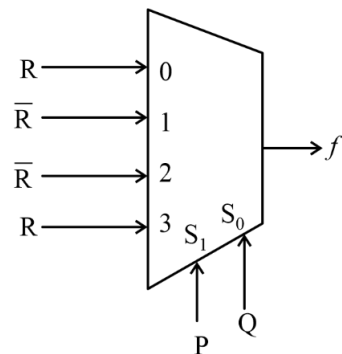
$f(x, y, a, b)$
 $\{$
 if (x is 1) $y = a$;
 else $y = b$;
 $\}$.

Which one of the following digital logic blocks is the most suitable for implementing this function?

- (a) Full adder (b) Priority encoder
 (c) Multiplexer (d) Flip-flop

8. [MCQ] [GATE-2010: 1M]

The Boolean expression of the output f of the multiplexer shown below is



- (a) $\overline{P \oplus Q \oplus R}$ (b) $P \oplus Q \oplus R$
 (c) $P + Q + R$ (d) $\overline{P + Q + R}$

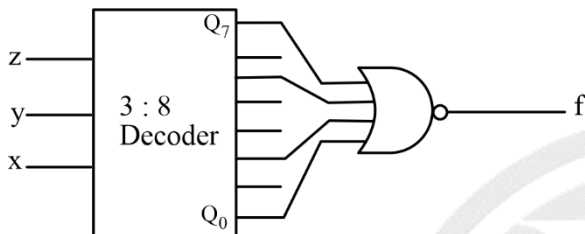
Decoder/De-MUX

9. [NAT] [GATE-2020: 1M]

If there are m input lines and n output lines for a decoder that is used to uniquely address a byte addressable 1 KB RAM, then the minimum value of $m + n$ is ____.

10. [MCQ] [GATE-2008: 1M]

What Boolean function does the circuit below realize?



- (a) $xz + \bar{x}\bar{z}$ (b) $x\bar{z} + \bar{x}z$
 (c) $\bar{x}\bar{y} + yz$ (d) $xy + \bar{y}\bar{z}$

Encoder and Priority Encoder

11. [MCQ] [GATE-2013: 1M]

In the following truth table, $V = 1$ if and only if the input is valid. What function does the truth table represent?

Inputs				Outputs		
D_0	D_1	D_2	D_3	X_0	X_1	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	x	1	0	1	0	1
X	x	x	1	1	1	1

What function does the truth table represent?

- (a) Priority encoder (b) Decoder
 (c) Multiplexer (d) Demultiplexer

ANSWER KEY

- | | | | |
|-----------|---------|---------|--------|
| 1. (-1) | 2. (c) | 3. (a) | 4. (5) |
| 5. (a) | 6. (d) | 7. (c) | 8. (b) |
| 9. (1034) | 10. (b) | 11. (a) | |

SOLUTIONS

1. (-1)

$$B = 11111111 \Rightarrow -1$$

$$00000001$$

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2. (c)

$$\text{Given function } f(ABC) = \bar{A} + \bar{A}\bar{C} + \bar{A}B\bar{C}$$

- First, we minimize the given function

$$f(ABC) =$$

$$\bar{A}(B + \bar{B})(C + \bar{C}) + \bar{A}(B + \bar{B})\bar{C} + \bar{A}B\bar{C}$$

$$f(ABC) =$$

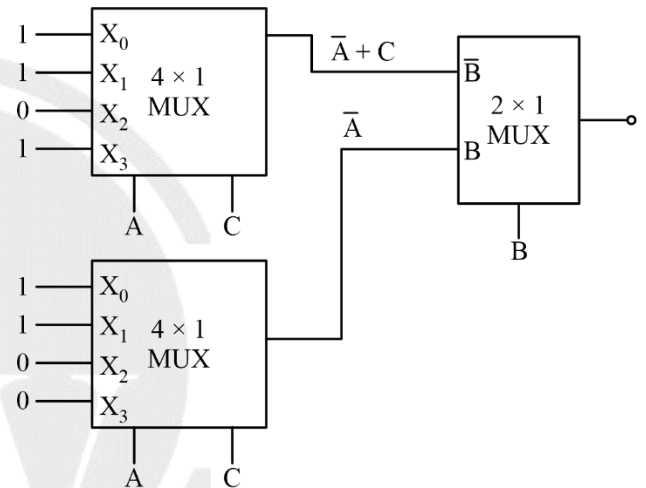
$$\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}C + \bar{A}BC + \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C}$$

$$f(ABC) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}C + \bar{A}BC + \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C}$$

$$f(ABC) = \sum m(0, 1, 2, 3, 5)$$

Now implementing on 2×1 mux.

	$\bar{B}$	B
$\bar{A}\bar{C}$	①	②
$\bar{A}C$	①	③
$A\bar{C}$	4	6
AC	⑤	7
	$\bar{A} + C$	$\bar{A}$



$$\therefore (X_0 X_1 X_2 X_3 X_4 X_5 X_6 X_7) = (11011100)$$

Hence correct answer is option (c).

3. (a)

$$\begin{aligned} F &= (\bar{X} + X)\bar{Y}\bar{Z} + (\bar{X} + X)\bar{Y}Z + XY\bar{Z} + \bar{X}YZ \\ &= \bar{X}\bar{Y}\bar{Z} + X\bar{Y}\bar{Z} + \bar{X}\bar{Y}Z + X\bar{Y}Z + XY\bar{Z} + \bar{X}YZ \\ &= \sum m(0, 1, 3, 4, 5, 6) \end{aligned}$$

4. (5)

Given:

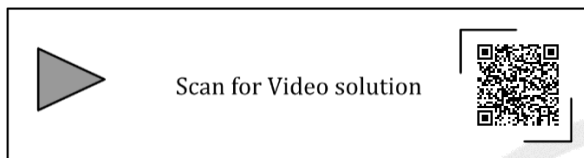
Number of registers = 32

Mux $\Rightarrow 32 \times 1$

Inputs = 32

Number of select line = $\log_2(\text{number of inputs})$

Number of select line = $\log_2 32 = 5$



5. (a)

$$F = \bar{P}Q + P\bar{Q}R + PQ\bar{R}$$

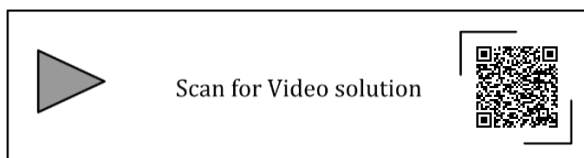
Standard canonical form

$$F = \bar{P}Q(\bar{R} + R) + P\bar{Q}R + PQ\bar{R}$$

$$= \bar{P}Q\bar{R} + \bar{P}QR + P\bar{Q}R + PQ\bar{R} = \sum m(2, 3, 5, 6)$$

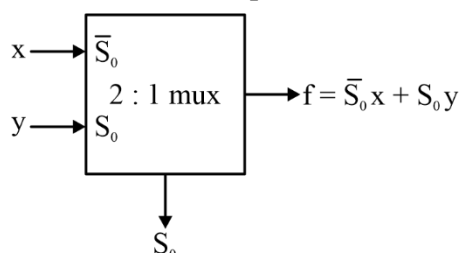
P \ QR	00	01	11	10
0			1	1
1		1		1

$$\bar{P}Q\bar{R} + \bar{P}QR + P\bar{Q}R + PQ\bar{R}$$



6. (d)

General 2:1 Mux expression



If S_0 is enabled and x, y are inputs then output

$$f = \bar{S}_0 x + S_0 y$$

Here in given problem output of MUX-1 is input to MUX -2

output MUX -1

$$f_1 = \bar{P}.0 + P.R = P.R$$

MUX -2 output

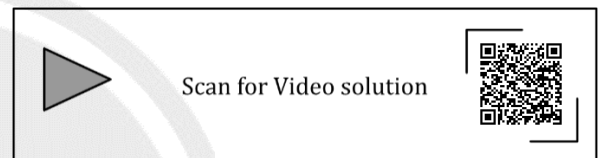
$$X = \bar{Q}. \bar{R} + Q. f_1$$

$$X = \bar{Q}. \bar{R} + Q.P.R$$

Its minimal sum of product form output

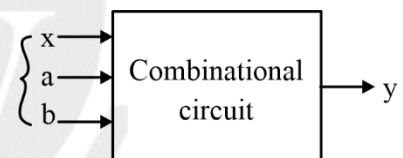
$$X = \bar{Q}. \bar{R} + PQR$$

Hence, correct answer is option (d).



7. (c)

Then given circuit will be as follows



From the given condition if

$$x = 1 \quad y = a$$

Otherwise, $y = b$

The truth table of the above circuit will be given as

x	a	b	y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

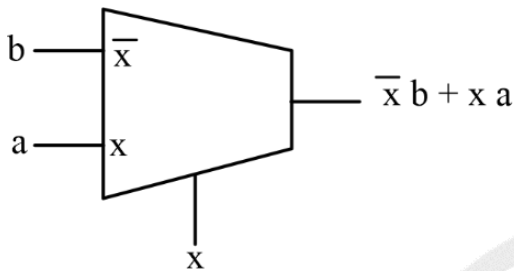
$$y(x, a, b) = \sum m(1, 3, 6, 7)$$

The k-map of the above function will be

		ab			
x	0	0 0	0 1	1 1	1 0
	1				

Annotations: $\bar{x}b$ (points to row x=0), xa (points to column a=1)

$$y = xa + \bar{x}b$$



Hence, correct answer is option (c).

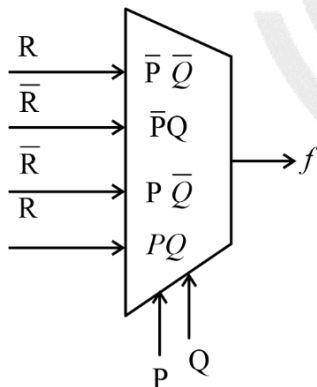


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8. (b)

Given multiplexer diagram redrawn as



The output of the given multiplexer will be given as

$$f = \bar{P}\bar{Q}R + \bar{P}Q\bar{R} + P\bar{Q}R + PQ\bar{R}$$

$$f = m_1 + m_2 + m_4 + m_7$$

$$f = \sum m(1, 2, 4, 7)$$

By using k-map we get

P	QR			
	0 0	0 1	1 1	1 0
0	0	1	0	1
1	1	0	1	0

$$f = \bar{P}\bar{Q}R + \bar{P}Q\bar{R} + P\bar{Q}R + PQ\bar{R}$$

$$f = P \oplus Q \oplus R$$

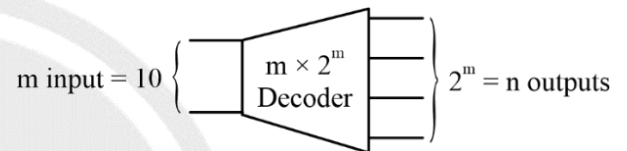
Hence, correct answer is option (b).



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9. (1034)



$$1 \text{ kB RAM} = 1024 = 2^{10}$$

$$m = 10$$

$$n = 2^{10} = 1024$$

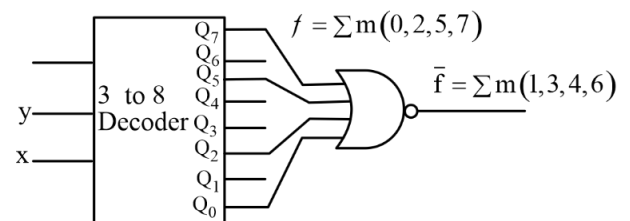
$$m + n = 10 + 1024 = 1034$$



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10. (b)



We can see that the given circuit is 3 to 8 decoder and its output will be given as.

$$\bar{f} = \sum m(1, 3, 4, 6)$$

By using k-map we get

		yz			
		00	01	11	10
x	0		1	1	
	1	1			1

$$f = \bar{x}z + x\bar{z}$$

Hence, correct answer is option (b).



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11. (a)

We know that

$D_0 \rightarrow \text{MSB}$

$D_3 \rightarrow \text{LSB}$

we can see that the given truth table is the truth table of priority encoder.

$V=0$ is invalid output.

$V=1$ is valid output

Hence, correct answer is option (a).



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CHAPTER

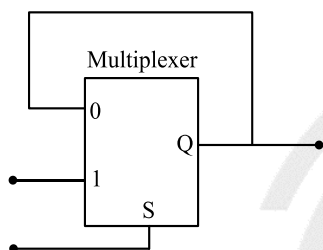
4

SEQUENTIAL CIRCUITS

Flip Flops

1. [MCQ] [GATE-2023: 1M]

The output of a 2-input multiplexer is connected back to one of its inputs as shown in the figure.

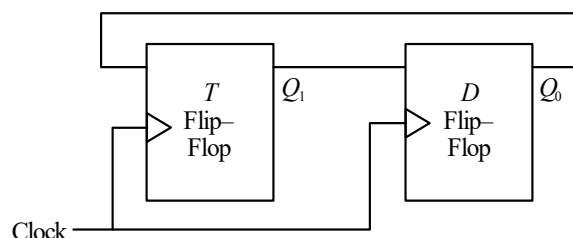


Match the function equivalence of this circuit to one of the following options.

- (a) D Flip-flop (b) D Latch
(c) Half-adder (d) Demultiplexer

2. [MCQ] [GATE-2017: 1M]

Consider a combination of T and D flip-flops connected as shown below. The output of the D flip-flop is connected to the input of the T flip-flop and the output of the T flip-flop is connected to the input of the D flip-flop.



Initially, both Q_0 and Q_1 are set to 1 (before the 1st clock cycle). The outputs

- (a) Q_1Q_0 after the 3rd cycle are 11 and after the 4th cycle are 00 respectively
(b) Q_1Q_0 after the 3rd cycle are 11 and after the 4th cycle are 01 respectively

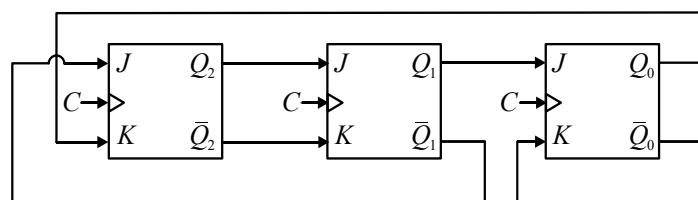
- (c) Q_1Q_0 after the 3rd cycle are 00 and after the 4th cycle are 11 respectively
(d) Q_1Q_0 after the 3rd cycle are 01 and after the 4th cycle are 01 respectively.

3. [MCQ] [GATE-2015: 1M]

A positive edge-triggered D flip-flop is connected to a positive edge-triggered JK flip-flop as follows. The Q output of the D flip-flop is connected to both the J and K inputs of the JK flip-flop, while the Q output of the JK flip-flop is connected to the input of the D flip-flop. Initially, the output of the D flip-flop is set to logic one and the output of the JK flip-flop is cleared. Which one of the following is the bit sequence (including the initial state) generated at the Q output of the JK flip-flop when the flip-flops are connected to a free-running common clock? Assume that $J = K = 1$ is the toggle mode and $J = K = 0$ is the state-holding mode of the JK flip-flop. Both the flip-flops have non-zero propagation delays.

- (a) 0110110... (b) 0100100...
(c) 011101110... (d) 011001100...

4. [MCQ] [GATE-2014: 1M]



The above synchronous sequential circuit built using JK flip-flops is initialized with $Q_2Q_1Q_0 = 000$. The state sequence for this circuit for the next 3 clock cycles is

- (a) 001, 010, 011 (b) 111, 110, 101
(c) 100, 110, 111 (d) 100, 011, 001

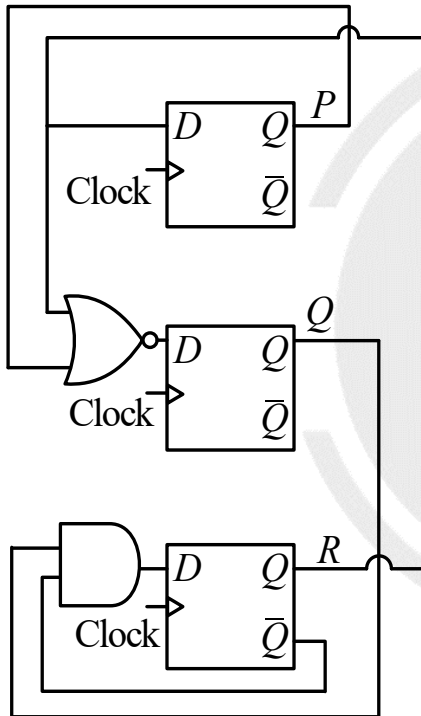
5. [MCQ] [GATE-2011: 1M]

The minimum number of D flip-flops needed to design a mod-258 counter is

- (a) 9 (b) 8
(c) 512 (d) 258

6. [MCQ] [GATE-2011: 1M]

Consider the following circuit involving three D-type flip-flops used in a certain type of counter configuration. If all the flip-flops were reset to 0 at power on, what is the total number of distinct outputs (states) represented by PQR generated by the counter?

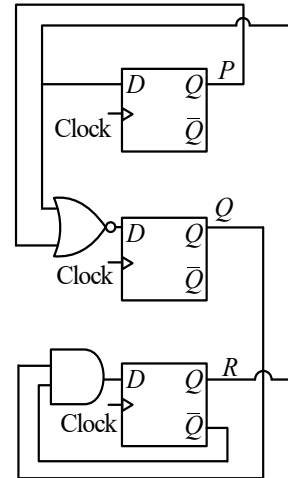


- (a) 3 (b) 4
(c) 5 (d) 6

7. [MCQ] [GATE-2011: 1M]

Consider the following circuit involving three D-type flip-flops used in a certain type of counter configuration.

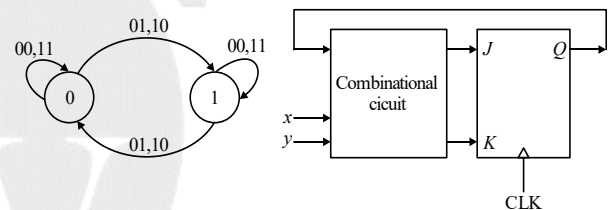
If at some instance prior to the occurrence of the clock edge, P, Q and R have a value 0, 1 and 0 respectively, what shall be the value of PQR after the clock edge?



- (a) 000 (b) 001
(c) 010 (d) 011

8. [MCQ] [GATE-2008: 1M]

Consider the following state diagram and its realization by a JK flip flop



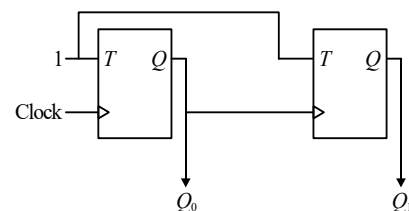
The combinational circuit generates J and K in terms of x, y and Q. The Boolean expressions for J and K are:

- (a) $(x \oplus y)'$ and $(x \oplus y)'$ (b) $(x \oplus y)'$ and $x \oplus y$
(c) $x \oplus y$ and $(x \oplus y)'$ (d) $x \oplus y$ and $x \oplus y$

Asynchronous Counter

9. [MCQ] [GATE-2010: 1M]

In the sequential circuit shown below, if the initial value of the output Q_1Q_0 is 00, what are the next four values of Q_1Q_0 ?

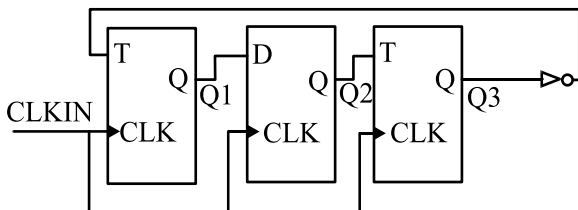


- (a) 11, 10, 01, 00 (b) 10, 11, 01, 00
(c) 10, 00, 01, 11 (d) 11, 10, 00, 01

Synchronous Counter

10. [MCQ] [GATE-2023: 2M]

Consider a sequential digital circuit consisting of T flip-flops and D flip-flops as shown in the figure. CLKIN is the clock input to the circuit. At the beginning, Q1, Q2 and Q3 have values 0, 1 and 1, respectively.

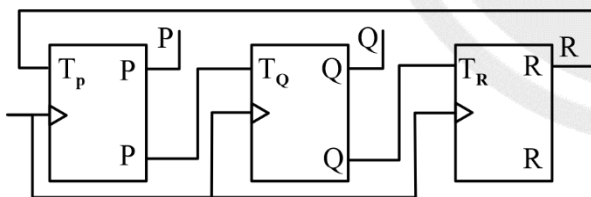


Which one of the given values of (Q1, Q2, Q3) can NEVER be obtained with this digital circuit?

- (a) (0, 0, 1) (b) (1, 0, 0)
(c) (1, 0, 1) (d) (1, 1, 1)

11. [MCQ] [GATE-2021: 1M]

Consider a 3-bit counter, designed using T flip-flops, as shown below.



Assuming the initial state of the counter given by PQR as 000. What are the next three states?

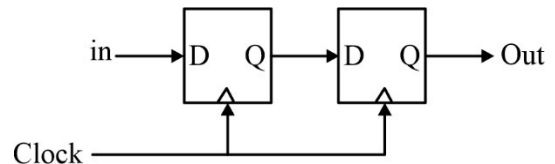
- (a) 011,101,000 (b) 001,010,111
(c) 011,101,111 (d) 001,010,000

12. [MCQ] [GATE-2021: 2M]

The minimum number of JK flip-flops required to construct a synchronous counter with the count sequence (0,0,1,1,2,2,3,3,0,0....)is_____.

13. [NAT] [GATE-2018: 1M]

Consider the sequential circuit shown in the figure, where both flip-flops used are positive edge-triggered D flip-flops.



The number of states in the state transition diagram of this circuit that have a transition back to the same state on some value of "in" is _____

14. [MCQ] [GATE-2017: 1M]

The next state table of a 2-bit saturating up-counter is given below.

Q ₁	Q ₀	Q ₁ ⁺	Q ₀ ⁺
0	0	0	1
0	1	1	0
1	0	1	1
1	1	1	1

The counter is built as a synchronous sequential circuit using T flip-flops.

The expressions for T₁ and T₀ are

- (a) $T_1 = Q_1 Q_0$, $T_0 = \overline{Q_1} \overline{Q_0}$
(b) $T_1 = \overline{Q_1} Q_0$, $T_0 = \overline{Q_1} + \overline{Q_0}$
(c) $T_1 = Q_1 + Q_0$, $T_0 = \overline{Q_1} + \overline{Q_0}$
(d) $T_1 = \overline{Q_1} Q_0$, $T_0 = Q_1 + Q_0$

15. [MCQ] [GATE-2015: 1M]

Consider a 4-bit Johnson counter an initial value of 0000. The counting sequence of this counter is-

- (a) 0,1,3,7,15,14,12,8,0
(b) 0,1,3,5,7,9,11,13,15,0
(c) 0,2,4,6,8,10,12,14,0
(d) 0,8,12,14,15,7,3,1,0

16. [MCQ]

[GATE-2014: 1M]

Let $k = 2^n$. A circuit is built by giving the output of an n -bit binary counter as input to an n -to- 2^n bit decoder.

This circuit is equivalent to a

- (a) k -bit binary up counter.
- (b) k -bit binary down counter.
- (c) k -bit ring counter.
- (d) k -bit Johnson counter.

17. [MCQ]

[GATE-2009: 1M]

How many $32K \times 1$ RAM chips are needed to provide a memory capacity of 256K-bytes?

- (a) 8
- (b) 32
- (c) 64
- (d) 128

Memory

18. [MCQ]

[GATE-2012: 00M]

The amount of ROM needed to implement a 4 bit multiplier is

- (a) 64 bits
- (b) 128 bits
- (c) 1 Kbits
- (d) 2 Kbits

□□□

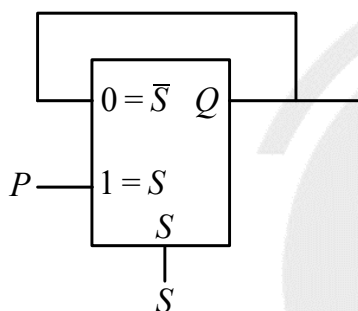


ANSWER KEY

- | | | | |
|---------|---------|---------|---------|
| 1. (b) | 2. (b) | 3. (a) | 4. (c) |
| 5. (a) | 6. (c) | 7. (d) | 8. (d) |
| 9. (a) | 10. (a) | 11. (a) | 12. (3) |
| 13. (2) | 14. (b) | 15. (d) | 16. (c) |
| 17. (c) | 18. (d) | | |

SOLUTIONS

1. (b)



$$Q_{n+1} = Q \bar{S} + DS$$

$$Q_{n+1} = Q.0 + D.1$$

$$Q_{n+1} = D$$

Q_{n+1} is generated after delay

Now in a flip flop we always consider edge but in the circuit, it is level sensitive, therefore the circuit is equivalent to a D-latch.



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2. (b)

Given present state of both flip flop before apply clock pulse is High (i.e. 1)

Clock	Q_1	Q_0
0	1	1
1	0	1
2	1	0
3	1	1
4	0	1
5	1	0
6	1	1

from above table the output state after 3rd and 4th clock pulse are 11 (Q_1Q_0) & 01 (Q_1Q_0) respectively hence correct option is (b).

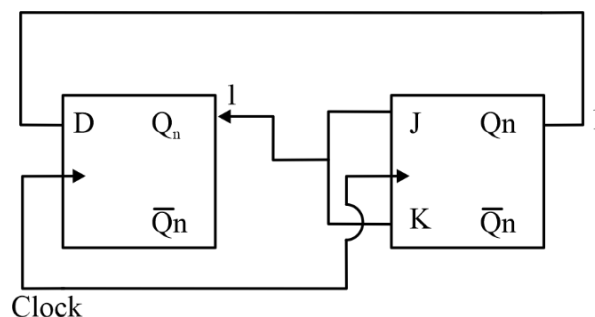


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3. (a)

According to the question, the given sequential circuit will look like as follows.

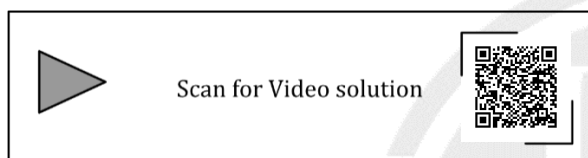


CLK	Q _n of JK Flip Flop
0	0
1	1
2	1
3	0
4	1
5	1

From the above truth table the Output of JK Flip Flop will be

$0 \rightarrow 1 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 1$
 $= 011011...$

Hence, correct answer is option (a).



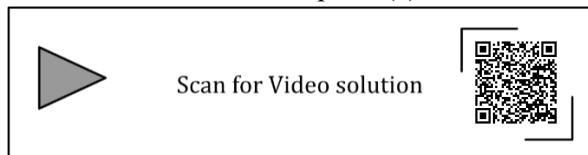
4. (c)

The truth table of the given flip-flop will be
 JK-flip flop

CLK	Q ₂	Q ₁	Q ₀
0	0	0	0
1	1	0	0
2	1	1	0
3	1	1	1

From the above truth table, the state sequence for next 3 clock cycles are 100, 110 and 111.

Hence, correct answer is option (c).



5. (a)

The number of Flip Flops needed to design Mod-M counter,

$$M \leq 2^n \quad \dots(i)$$

where n = number of flip flops

to solve (i) by taking log on both sides we get,

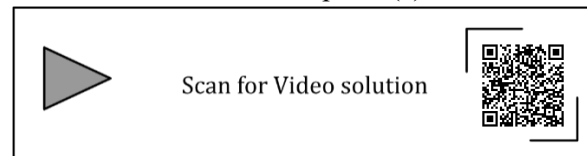
$$n \geq \log_2 M$$

$$n \geq \log_2 258 \quad [\because M = 258]$$

$$n \geq 8.01122$$

$$n \cong 9$$

Hence, correct answer is option (a).



6. (c)

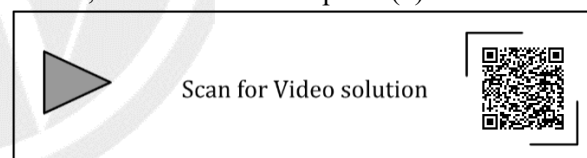
As per the given conditions, the truth table will be given as

CLOCK	P	Q	R
0	0	0	0
1	0	1	0
2	0	1	1
3	1	0	0
4	0	0	0

We can see that the above truth table is the output of a mod- 4 counter.

Hence total number of distinct states = 4

Hence, correct answer is option (b)



7. (d)

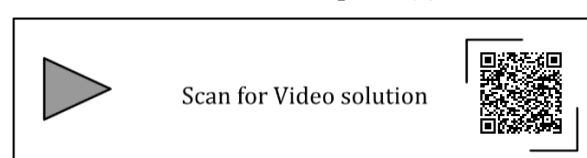
In the circuit PQR = 010

Before applying the clock, input of the Flip-flop = 011

So, in the D-FF whatever the inputs applied that will come at the output along with clock.

So, output of counter = 011

Hence, correct answer is option (d).



8. (d)

We know that truth table of JK flip-flop is given by

Q_n	Q_{n+1}	J	K
0	1	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Now the truth table of given circuit is given by

x	y	Q_n	Q_{n+1}	J	K
0	0	0	0	0	×
0	0	1	1	×	0
0	1	0	1	1	×
0	1	1	0	×	1
1	0	0	1	1	×
1	0	1	0	×	1
1	1	0	0	0	×
1	1	1	1	×	0

Solving J & K using k-map, we get

Q_n x \ y				
	00	01	11	10
0	0	×	×	1
1	1	×	×	

$$J = x\bar{y} + \bar{x}y = x \oplus y$$

Q_n x \ y				
	00	01	11	10
0	×		1	×
1	×	1		×

$$K = \bar{x}y + x\bar{y} = x \oplus y$$

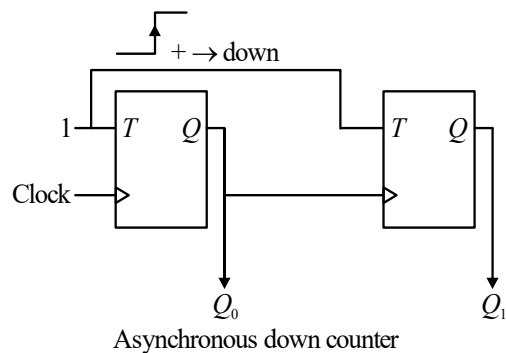
Hence, correct answer is option (d).



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9. (a)



The truth table of the above sequential circuit is given as follows.

CLK	Q_1	Q_0
0	0	0
1	1	1
2	1	0
3	0	1
4	0	0

From the above truth table, we can see that it is an output of an Asynchronous down counter.

Hence, correct answer is option (a).



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10. (a)

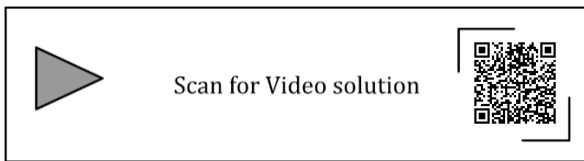
The characteristics table of given sequential circuit

Clock	$T = \bar{Q}_3$	$D = Q_1$	$T = Q_2$	Q_1	Q_2	Q_3
0	-	-	-	0	1	1
1	0	0	1	0	0	0
2	1	0	0	1	0	0
3	1	1	0	0	1	0
4	1	0	1	1	0	1

5	0	1	0	1	1	1
6	0	1	1	1	1	0
7	1	1	1	0	1	1

From above table states involved 011, 000, 100, 010, 101, 111, 110. The only state 001 can never be achieved.

Hence correct answer is option (a).



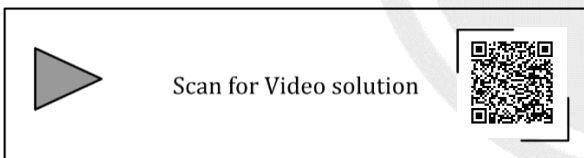
11. (a)

According to question, the initial state of counter is PQR = 000

CLK	P	Q	R
0	0	0	0
1	0	1	1
2	1	0	1
3	0	0	0

From the truth table the next three states are

011 → 101 → 000



12.. (3)

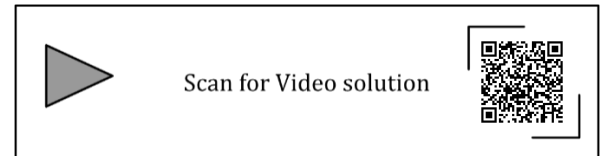
A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1
0	0	0
0	0	1
0	1	0
0	1	1

If we ignore the value of 'C'

00 → 00 → 01 → 01 → 10 → 10 → 11 → 11
 0 → 0 → 1 → 1 → 2 → 2 → 3 → 3 → 0 → 0

Design the up counter of 3bit ignore the LSB.

So, numebr of FF's are = 3

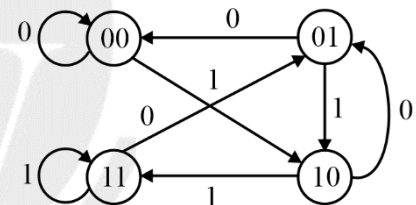


13. (2)

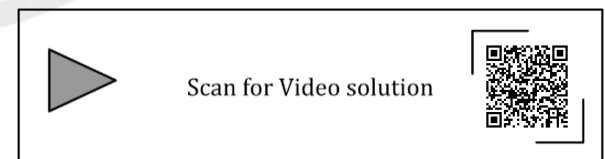
The characteristics table of given Flip-Flop

Input	D ₁	D ₂	Q ₁ ⁺	Q ₂ ⁺ (Out)
0	0	0	0	0
1	0	0	1	0
0	0	1	0	0
1	0	1	1	0
0	1	0	0	1
1	1	0	1	1
0	1	1	0	1
1	1	1	1	1

From above characteristics table state transition diagram of the given positive edge-triggered D flip flop is



From above state transition diagram after 2 state it come back to some value.



14. (b)

Q ₁	Q ₀	Q ⁺	Q ₀ ⁺	T ₁	T ₀
0	0	0	1	0	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	1	1	0	0

For T₀

$Q_1 \backslash Q_0$	0	1
0	1	1
1	1	0

$$T_0 = \bar{Q}_1 + \bar{Q}_0$$

For T_1

$Q_1 \backslash Q_0$	0	1
0	0	1
1	0	0

$$T_1 = \bar{Q}_1 Q_0$$

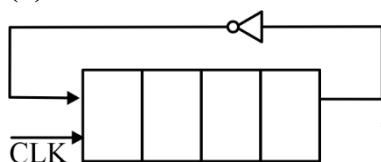
Hence correct answer is option (b).



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15. (d)



0 0 0 0 → 0

1 0 0 0 → 8

1 1 0 0 → 12

1 1 1 0 → 14

1 1 1 1 → 15

0 1 1 1 → 7

0 0 1 1 → 3

0 0 0 1 → 1

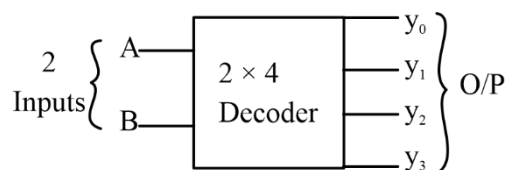


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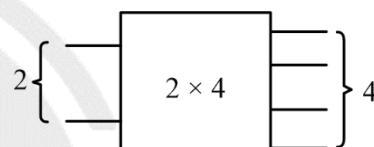
16. (c)

Let 2×4 Decoder as shown below, be at the output

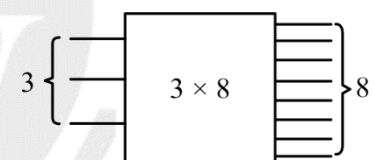


Now, according to question 2-bit binary counter is connected in the input of the above Decoder. The output of the combination will be given as

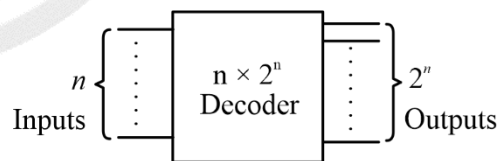
A	B	y ₀	y ₁	y ₂	y ₃
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1



4 bit Ring counter



8 bit Ring counter



2^n bit Ring counter
k-bit Ring counter

Ring counter of 2^n bits i.e. k-bit ring counter.

Hence, correct answer is option (c)



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17. (c)

Number of required chips for memory capacity

$$= 256 \times 2^{10} \times 8$$

Number of available chips for RAM capacity

$$= 32 \times 2^{10} \times 1$$

Now,

Number of chips needed to build 256K-bytes will be

$$\text{Number of chips} = \frac{\text{Required}}{\text{Available}} = \frac{256 \times 2^{10} \times 8}{32 \times 2^{10}} = 64$$

Hence, correct answer is option (c).



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18. (d)

$$\begin{array}{l} \boxed{4} \quad A = A_3 A_2 A_1 A_0 \rightarrow 2^4 \\ \boxed{4} \quad B = B_3 B_2 B_1 B_0 \rightarrow 2^4 \\ \hline \Rightarrow \underline{M_7 M_6 M_5 M_4 M_3 M_2 M_1 M_0} \\ \rightarrow 8 \text{ bits} \quad 8 \text{ o/p} \end{array}$$

$$\begin{aligned} \text{ROM} &= \overbrace{2^4 \times 2^4}^{\text{Input}} \times \underbrace{2}_{\text{output}} \\ &= 2^8 \times 2^3 \\ &= 2^{11} \\ &= 2 \times 2^{10} \\ &= 2 \text{ k bits} \end{aligned}$$



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